Dynamics of Quantum Mechanical Particles in Exotic Nanostructures

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We present a study of the dynamics of quantum mechanical particles confined in exotic nanostructures. The energy eigenstates and probability density functions are evaluated. Wherever possible, analytic solutions are utilised. For non-separable cases, a variational approach was used. In particular, we explain the emergence of pseudo-potentials from confinement effects in non-trivial geometries and the resulting consequences for particle dynamics. The result is a *naive* version of Quantum Gravity. It is then shown that all 2D solutions can be recovered as degeneracies of a single 3D case. Possible applications of quantum dot arrays in memory storage systems and methods to exploit favourable optical and electrical properties are also discussed.

I. INTRODUCTION

This project explores the consequences of the confinement of quantum mechanical particles in nanostructures with exotic geometries. All the cases explored are for a free particle with the boundary conditions dictating the geometry of the nanostructure. What result are effectively curved infinite square well potentials with periodic boundary conditions (except for the linear strip) similar to the Born-von Karman case. Whenever possible, the separability of the solutions was exploited and analytic solutions were obtained. However, for the non-trivial cases where the differential equations were not separable, an alternative variational approach was utilised. The curved geometries produced some extremely interesting effects reminiscent of the naive attempts at quantum gravity during the early 1950s. Although this was never an intended aim of the project, our results can also be viewed as an insight into the consequences of a simple attempt at recovering quantum gravity from the solutions of Schrödinger's equation in curved spaces.

The cases explored follow a natural progression in terms of complexity in the geometry. The first case studied is that of a linear strip as a contrasting preamble to the curved nanostructures. The results obtained provide a useful insight into the cases that follow. The second case studied was that of a circular annulus. Even with this relatively simple case, the effects of curvature were immediately apparent where we obtained a *negative energy* term that had an effect similar to an attractive confining potential. We refer to this later as a pseudo-potential as this is not an externally enforced potential resulting from electromagnetic effects as is usually the case. It was, in fact, purely the result of the curvature intrinsic to our chosen geometries.

After the circular annulus, the solution for the spherical shell was evaluated to investigate how the pseudo potential extended to 3D space. The potential term however disappears in this case. The mathematical explanation for this is harder to obtain if the geometries are considered to be embedded in higher dimensional spaces. What this means is that although from an extrinsic viewpoint the spherical shell appears to be defined in 3D space, if the radii defining each of the boundaries are considered fixed, then the space effectively reduces to a 2D geometry with the angular displacement coordinates being the only allowed degrees of freedom, thereby reducing the problem to an intrinsically 2D case. This idea can also be easily extended to the circular annulus by considering it to be effectively a *squashed cylinder*. This approach, in fact, also alludes to why we are able to recover all the 2D cases as degeneracies of the (elliptic) helical annulus.

The final two cases were those of the elliptical annulus and the helical annulus for which the spirals were kept elliptical to make the solution as general as possible. Both these had to be solved numerically using iterative variational methods.

Some of the possible applications are also outlined with an emphasis on how the aforementioned nanostructures, and particularly the curvature effects can be utilised in various areas of Solid State research.

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II. LINEAR STRIP

The linear strip seemed like a natural starting point for exploring 2D nanostructures. It was important in bringing to light not only how important the effects of curvature were, but also how the wavefunctions for the curved nanostructures could intuitively be understood as *bent* versions of those for the linear strip.

A. Eigenfunctions

The natural choice was to opt for cartesian coordinates for the solution of this problem. The linear strip was defined as having dimensions of length a along the x axis starting from $x = 0$, and similarly width b along the y axis. The wavefunctions were made to vanish at the ends by enforcing that $\psi(0, y) = \psi(a, y) = \psi(x, 0) = \psi(x, b) = 0$. We consequently obtained the Time Independent Schrödinger Equation (TISE) as follows,

$$\frac{-\hbar^2}{2m_0}\nabla^2\psi = \frac{-\hbar^2}{2m_0}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2}\right) = E\psi \tag{A.1}$$

This can be solved via separation of variables by defining a wavefunction $\psi(x, y) = X(x)Y(y)$ which leads to a normalised solution (see Appendix A-1) as follows,

$$\psi_{m,n}(x, y) = \sqrt{\frac{4}{ab}}\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) \tag{A.2}$$

The figure below shows the (a) (1,1), (b) (2,1), (c) (1,2) and (d) (2,2) eigenfunctions for the linear strip with length $a = 2$ and width $b = 1$. The effect of varying m and n is immediately obvious as increasing m or n or both requires more oscillations to be fit inside the boundaries resulting in more cells along the x and y axes respectively. The elliptical contours are a consequence of the difference in length along the x and y directions. This is evident from the fact that for the (2,1) eigenfunction where the linear strip is effectively reduced to two square cells, the resulting contours are circular in shape. However, for the other three cases where there is a disparity in linear dimensions along the two axes, we end up with elliptical contours instead. Note also that the probability density functions $|\psi_{m,n}|^2$ would be very similar to the wavefunctions. It is interesting

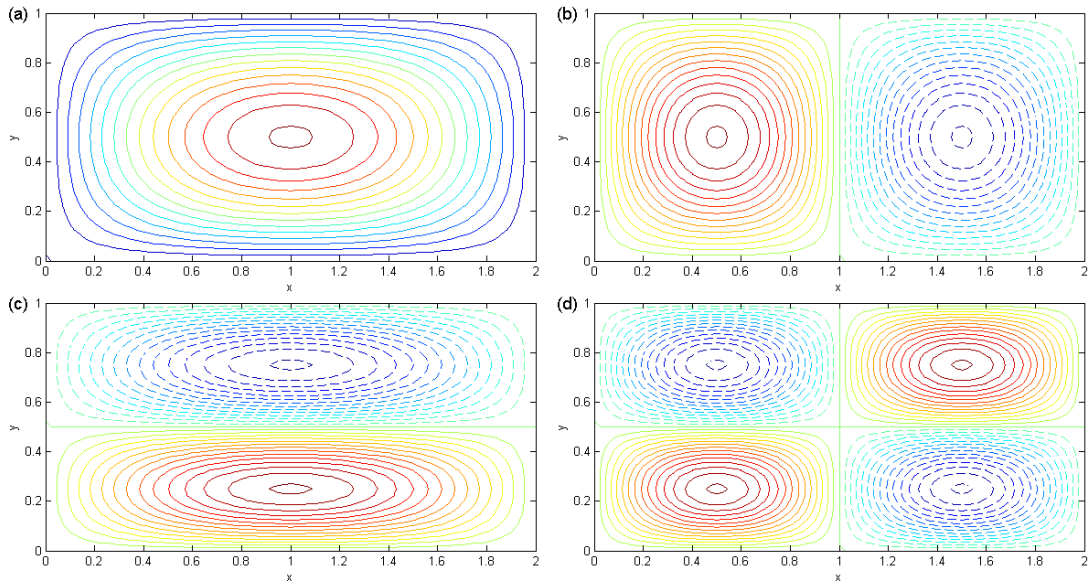

FIG. 1: Eigenfunctions $\psi_{m,n}$ of a linear strip of length $a = 2$ and width $b = 1$ for quantum numbers (m, n): (a) (1,1), (b) (2,1), (c) (1,2) and (d) (2,2). The dashed lines indicate negative values.

to note how closely the shape of the eigenfunctions (and probability density functions) resembles the ubiquitous structure of the palisade cells in the leaves of green plants. This may perhaps be linked to how in a similar manner the wavefunction of the matter present inside these cells has to obey the boundary conditions enforced by the cellulose cell walls.

The quantum numbers m and n are integers as dictated by the boundary conditions. This then leads to energy eigenvalues which are defined as,

$$E_{m,n} = \frac{\hbar^2}{2m_0}\left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right] \tag{A.3}$$

This is of course the same as the result for a 2D infinite square well with a and b defining the dimensions of the well along the x and y axes respectively. We will repeatedly encounter remnants of this solution for the other 2D nanostructures which are essentially obtained by *bending* and *twisting* the linear strip.

B. Symmetry

We must now analyse how the eigenfunctions transform under the symmetry operations of the *rectangular* strip:

- E - identity,
- $C_2(z)$ - rotation by π about $(a/2, b/2, z)$ axis,
- $C_2(y)$ - rotation by π about $(a/2, y, 0)$ axis,
- $C_2(x)$ - rotation by π about $(x, b/2, 0)$ axis,

which together belong to the group D_2.

Under an active transformation $R_d\psi(x,y)$, the bases transform as $R_d\psi(x,y) = \psi\Big(R_d^{-1}(x,y)\Big)$, $\forall d \in D_2$ [1],[2]. This therefore yields:

- $R_E\psi(x,y) = \psi(x,y)$,
- $R_{C_2(z)}\psi(x,y) = (-1)^{m+n}\psi$,
- $R_{C_2(y)}\psi = (-1)^{m+1}\psi$,
- $R_{C_2(x)}\psi = (-1)^{n+1}\psi$,

where we can immediately see that the wavenumbers m and n dictate the parity of the eigenstates.

The table below summarises the symmetry properties for the eigenstates of the linear strip,

D_2	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	Basis	Eigenstate
A	1	1	1	1	$-$	$\Gamma_{odd,odd}$
B_1	1	1	-1	-1	z, R_z	$\Gamma_{even,even}$
B_2	1	-1	1	-1	y, R_y	$\Gamma_{odd,even}$
B_3	1	-1	-1	1	x, R_x	$\Gamma_{even,odd}$

TABLE I: Character table for the D_2 symmetry group of the linear strip.

Note that since D_2 is an *abelian group* with exclusively 1D irreducible representations, there is no normal degeneracy. Any degeneracy in the eigenstates is purely accidental. In the case of a square ($a = b$), however, the states do exhibit normal degeneracy since the symmetry group is no longer D_2. The symmetry group for a square is the *non-abelian* group D_4 instead which has 2D irreducible representations, and consequently exhibits normal degeneracy.

III. CIRCULAR ANNULUS

The next case that we investigated was that of the circular annulus. This was perhaps the most important from the point of view of the direction of the project. The negative energy term that we obtained in the thin shell energy expansion became the overarching focus of the project. A significant amount of effort was dedicated to understanding and evaluating this single term, and to predicting how this extended to a general curved surface. From a *logistical viewpoint*, the circular annulus is effectively the result of taking the linear strip and joining its ends while ensuring that the mechanical stress along the strip is independent of the angular position. The boundary conditions enforced also ensure the continuity of the wavefunction and enforce the symmetry of the circle.

A. Eigenfunctions

The solution for the eigenfunctions of the circular annulus progresses in a manner similar to that for the linear strip, the only differences being in the choice of coordinates and boundary conditions. Polar coordinates (r, ϕ) were used in this case to exploit the circular symmetry [3]. The boundary conditions too were chosen to emphasise this symmetry. Consequently, along the radial direction we had $\psi(a, \phi) = \psi(b, \phi) = 0$ where a and b are the inner and outer radii respectively. Along the azimuthal direction, we had $\psi(r, \phi) = \psi(r, \phi + 2\pi)$.

Therefore, we obtain the TISE,

$$-\frac{\hbar^2}{2m_0}\nabla^2\psi = -\frac{\hbar^2}{2m_0}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\psi}{\partial\phi^2}\right] = E\psi \tag{A.1}$$

This can then be solved via separation of variables by defining a wavefunction $\psi(r, \phi) = R(r)\Phi(\phi)$ which leads to the following form for the eigenstates (see Appendix A-2),

$$\psi_{m,n}(r, \phi) = C\Big[Y_m(k_{m,n}a)J_m(k_{m,n}r) - J_m(k_{m,n}a)Y_m(k_{m,n}r)\Big]e^{im\phi} \tag{A.2}$$

where m and n are the azimuthal and radial wavenumbers respectively. C is a normalisation constant that now depends on the specific choice of wavenumbers. J_m and Y_m are bessel functions of order m of the first and second kind respectively. The bessel functions J_m and Y_m have eigenvalues $k_{m,n} = \sqrt{2m_0E_{m,n}}/\hbar$.

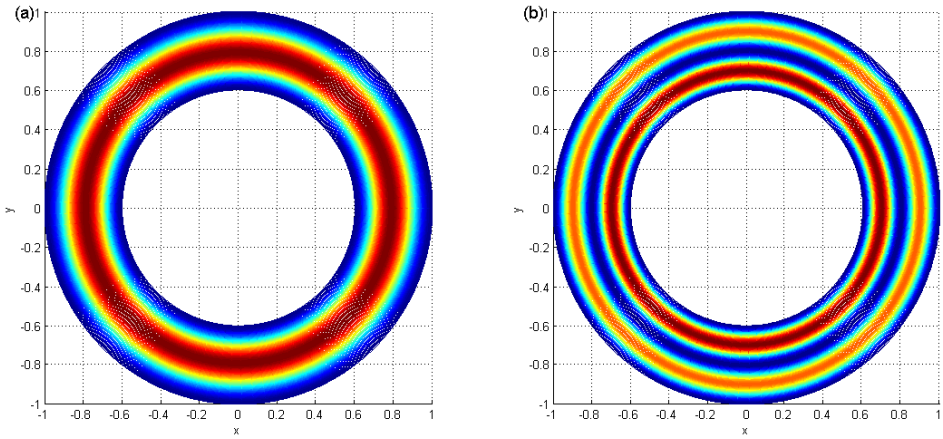

FIG. 2: Probability density functions $|\psi_{m,n}|^2$ for a circular annulus of inner radius $a = 0.6$ and outer radius $b = 1$, with quantum numbers (m, n): (a) $(m, 1)$ and (b) $(m, 2)$. Note that we can use any value for m here since the symmetry of the annulus makes the probability density independent of angular position. This should be obvious since the angular uniformity of the circle results in all angular positions being equally probable, thereby making $|\psi_{m,n}|^2$ independent of m.

The figure above shows the probability density functions $|\psi_{m,n}|^2$ for a circular annulus of inner radius $a = 0.6$ and outer radius $b = 1$, with quantum numbers (m, n): (a) $(m,\ 1)$ and (b) $(m,\ 2)$. The circular symmetry of the annulus makes $|\psi_{m,n}|^2$ independent of m. Consequently only varying the radial wavenumber n produces any physical consequences. The trend is easy to understand - increasing n increases the number of *coloured rings* inside the annulus since we have to accommodate an increasing number of *oscillations* along the radial direction which has an infinite square well boundary condition. This, as predicted is the result of uniformly *bending* the linear strip while ensuring that the mechanical stress is kept uniform throughout. It is in fact, this uniformity in mechanical stress which manifests in the circular shape being obtained as a consequence of which $|\psi_{m,n}|^2$ is independent of m.

In the thin shell limit, i.e. $b/a - 1 = \delta \ll 1$, the energy eigenvalues (see Appendix A-2) are,

$$E_{m,n} = \frac{\hbar^2 k_{m,n}^2}{2m_0} = \frac{\hbar^2}{2m_0}\left[\left(\frac{n\pi}{a\delta}\right)^2 + \left(\frac{m}{a}\right)^2 - \left(\frac{1}{2a}\right)^2\right] + O(\delta) \tag{A.3}$$

The first term is the radial confinement energy which is a remnant of the square well character where $\frac{\hbar^2}{2m_0}\left(\frac{n\pi}{a\delta}\right)^2$ is exactly the energy for an infinite square well of width $a\delta$. The second term $\frac{\hbar^2}{2m_0}\left(\frac{m}{a}\right)^2$ results from the azimuthal motion. However, the most important and most peculiar term by far is the third term $-\frac{\hbar^2}{2m_0}\left(\frac{1}{2a}\right)^2$. The negative sign here indicates that this is a *confinement* term that limits the kinetic energy of the particle. It is effectively a *pseudo potential energy* since there is no physical attractive force present to *attract* the particle inward. As we will see in the next subsection, this term is purely a consequence of the curvature intrinsic to our space. What follows now is an outline of the origin of this term. We will subsequently justify the emergence of this term and derive a general expression that will allow us to compute this term in general.

B. Origin of the Pseudo-potential

We shall now obtain a mathematical justification for the emergence of the pseudo-potential. Later, we also provide a possible physical interpretation for this to provide greater insight into the dynamics of this process. It is worth reiterating that there is by no means any *external agent*, e.g. an electric field (or any source of a potential) present to impose this condition. As we will derive, this term is purely a consequence of the innate curvature of our system. It is therefore considerably more fruitful to approach this problem by using an intrinsic geometric approach rather than considering our manifold to be embedded in a higher dimensional space. Consequently for any surface embedded in 3D space, we are able to reduce our degrees of freedom down to effectively two.

We begin by considering the motion of a particle constrained to move on a surface S which is sandwiched between two surfaces parallel to it [5]. The *roof* and *floor* of our system are separated by a distance d with S being equidistant from the *roof* and the *floor* at every point. The figure below is a side-view of our *sandwich*.

We can use the orthogonal coordinates x^i, $\forall i \in \{1, 2\}$; to span our surface S. Using the normal vector $\hat{\mathbf{n}} = \hat{\mathbf{n}}(x^i)$ for S, we can transition to surfaces parallel to S at every point. Consequently, any point $\mathbf{r}$ on our surfaces can be defined as $\mathbf{r} = \mathbf{a}(x^i) + \hat{\mathbf{n}}\tau$. We can define $\tau = x^3$ as a third

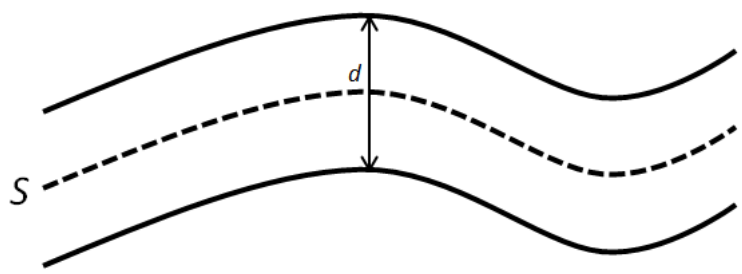

FIG. 3: Side-view of our parallel surfaces with S sandwiched in the middle.

orthogonal coordinate which allows us to move to parallel surfaces. Consequently, our total set of orthogonal coordinates is defined by x^μ, $\forall \mu \in \{1,2,3\}$. Differentiating $\mathbf{r}$, we obtain,

$$d\mathbf{r} = \frac{\partial \mathbf{a}}{\partial x^1}dx^1 + \frac{\partial \mathbf{a}}{\partial x^2}dx^2 + \tau\frac{\partial \hat{\mathbf{n}}}{\partial x^1}dx^1 + \tau\frac{\partial \hat{\mathbf{n}}}{\partial x^2}dx^2 + \hat{\mathbf{n}}d\tau$$

$$= \left(\frac{\partial \mathbf{a}}{\partial x^1} + \tau\frac{\partial \hat{\mathbf{n}}}{\partial x^1}\right)dx^1 + \left(\frac{\partial \mathbf{a}}{\partial x^2} + \tau\frac{\partial \hat{\mathbf{n}}}{\partial x^2}\right)dx^2 + \hat{\mathbf{n}}d\tau \tag{B.1}$$

This therefore gives us the metric,

$$g_{\mu\nu} = \begin{pmatrix} \left(\frac{\partial \mathbf{a}}{\partial x^1} + \tau\frac{\partial \hat{\mathbf{n}}}{\partial x^1}\right)^2 & 0 & 0 \\ 0 & \left(\frac{\partial \mathbf{a}}{\partial x^2} + \tau\frac{\partial \hat{\mathbf{n}}}{\partial x^2}\right)^2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{B.2}$$

where we used the fact that the tangent vectors in an orthogonal coordinate system would also be mutually orthogonal at every point, to obtain our metric.

Rodrigues' curvature formula allows us to relate the normal vector $\hat{\mathbf{n}}$ of a surface S with surface function $\mathbf{a}$, to the principle curvatures k_1 and k_2, which are simply reciprocals of the principle radii of curvature R_1 and R_2 respectively,

$$d\hat{\mathbf{n}} + k_i d\mathbf{a} = 0, \quad i \in \{1,2\} \tag{B.3}$$

Differentiating with respect to x^i,

$$\frac{\partial \hat{\mathbf{n}}}{\partial x^i} = -\frac{1}{R_i}\frac{\partial \mathbf{a}}{\partial x^i} \tag{B.4}$$

and substituting into the expression for the metric, we obtain

$$g_{\mu\nu} = \begin{pmatrix} \left(1 - \frac{\tau}{R_1}\right)^2 \left|\frac{\partial \mathbf{a}}{\partial x^1}\right|^2 & 0 & 0 \\ 0 & \left(1 - \frac{\tau}{R_2}\right)^2 \left|\frac{\partial \mathbf{a}}{\partial x^2}\right|^2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{B.5}$$

where we can relate the determinant of the metric on S, defined as g_0 to the metric on any surface displaced from S by a distance τ as,

$$g = g_0\gamma \tag{B.6}$$

where,

$$\gamma = \left(1 - \frac{\tau}{R_1}\right)^2\left(1 - \frac{\tau}{R_2}\right)^2 \tag{B.7}$$

We can now use this expression for the metric to solve the TISE for a general Riemannian manifold. The Laplace-Beltrami operator then is,

$$\nabla^2 = g^{-1/2}\partial_\mu(g^{1/2}g^{\mu\nu}\partial_\nu) \tag{B.8}$$

where g is simply the determinant of the metric $g_{\mu\nu}$ with the inverse metric denoted by $g^{\mu\nu}$. Separating out the $\tau = x^3$ component, we obtain,

$$\nabla^2 = g^{-1/2}\Big[\partial_i(g^{1/2}\eta^{ij}\partial_j) + \partial_\tau(g^{1/2}g^{\tau\tau}\partial_\tau)\Big] \tag{B.9}$$

where we can define $g_{\mu\nu} = \eta_{ij} \oplus g_{\tau\tau}$ since the metric is a diagonal matrix with the elements of the inverse metric given by the reciprocals of the corresponding elements of the metric matrix. Note also that η_{ij} is now the metric over the surface S. We finally arrive at our TISE,

$$-\frac{\hbar^2}{2m_0}\Big[g^{-1/2}\partial_i g^{1/2}\eta^{ij}\partial_j\psi + g^{-1/2}\partial_\tau g^{1/2}\partial_\tau\psi\Big] = E\psi \tag{B.10}$$

Substituting $\psi = \gamma^{-1/4}\chi$ [5], the second term in the TISE becomes (see Appendix B),

$$g^{-1/2}\partial_\tau g^{1/2}\partial_\tau \psi = \gamma^{-1/4}(\partial_\tau^2 \chi + U\chi) \tag{B.11}$$

where,

$$U = -\frac{1}{16}(\partial_\tau \ln\gamma)^2 - \frac{1}{4}\partial_\tau^2 \ln\gamma \tag{B.12}$$

The full TISE then is,

$$-\frac{\hbar^2}{2m_0}\Big[g^{-1/2}\partial_i g^{1/2}\eta^{ij}\partial_j \gamma^{-1/4}\chi + \gamma^{-1/4}\partial_\tau^2\chi + \gamma^{-1/4}U\chi)\Big] = E\gamma^{-1/4}\chi \tag{B.13}$$

where we can divide out the coordinate independent term $\gamma^{-1/4}$. We can therefore compactly write our TISE as,

$$-\frac{\hbar^2}{2m_0}\Big[\nabla_i^2\chi + \partial_\tau^2\chi + U\chi\Big] = E\chi \tag{B.14}$$

where we used the fact that $\chi_\mu(x^\mu) = \chi_i(x^i)\chi_\tau(\tau)$ since the coordinates are mutually independent and orthogonal. Note that the $-\frac{\hbar^2}{2m_0}\nabla_i^2\chi$ term gives the kinetic energy for the motion on S, where as the $-\frac{\hbar^2}{2m_0}\partial_\tau^2\chi$ term is responsible for the energy of motion along the $\hat{\mathbf{n}}$ direction.

The final term is in fact our *pseudo potential energy* $-\frac{\hbar^2}{2m_0}U\chi$ (see Appendix B),

$$U = \frac{1}{4}\left(\frac{1}{R_1 - \tau} - \frac{1}{R_2 - \tau}\right)^2 \tag{B.15}$$

where,

$$\lim_{\tau\to 0} U = \frac{1}{4}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)^2 = \frac{1}{4}\Big(K_1 - K_2\Big)^2 \tag{B.16}$$

This, as promised, is the general expression for the *pseudo-potential*. It is very simply a function of the principle curvatures of the surface. We can now verify this for the case of the circular annulus, which as mentioned before can be extended to 3D space by considering it to be a *squashed cylindrical shell*. Thus, in this case where our cylindrical shell is *thin*, $R_1 = a$ is the radius of the annular ring itself and R_2 is the curvature radius which extends along the z axis. Since R_2 extends indefinitely along the z axis, it is therefore equal to ∞. Using this, we can calculate U for the circular annulus,

$$U = \frac{1}{4}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)^2 = \frac{1}{4}\left(\frac{1}{a} - \frac{1}{\infty}\right)^2 = \frac{1}{4a^2} \tag{B.17}$$

Note that it should be immediately obvious that U is constant since both R_1 and R_2 are constant. We can therefore also predict that for a surface for which the radii of curvature vary with position, so does U. This is of course the case for an ellipse. It is important to note that this is a purely quantum mechanical effect as solving this problem for the classical analogue of the TISE, i.e. the diffusion equation, we end up with no such term [3]. This is because in the classical limit, the Lagrangian of the system inherently incorporates the boundary constraints of the system through the use of normal coordinates. Consequently we get no additional terms that demonstrate a Lagrangian dependence on the local geometry. Indeed, in a *spooky* sense, ubiquitous to quantum mechanics, the particle demonstrates knowledge of the global properties of the system. This effect has been accounted for in several other publications which in a similar manner study the effects of curvature on the particle dynamics [6], [7], [8], [9], [10], [11], [12].

Perhaps a nice way to physically understand this effect is by using an analogy [6] where we focus on the motion of a race car driver around a track. Considering first the case of the linear strip which now defines our *race track*, we notice that the freedom to swerve sideways and simultaneously

move down the track both contribute to the kinetic energy. This of course is what we expect in both the classical and quantum cases for the linear strip. However, now if we join the two ends of the linear track to form a circular race track, the same race car driver now has to slam on the brakes to ensure that he doesn't crash into the sidewalls of the track. He notices that although the road is still the same, he has to steer around more consistently. Consequently, in order to maintain his motion within the confines of the track, the driver has to effectively lower his energy. This is thus the physical origin of our *pseudo-potential* term. However, it must be noted that this is merely for illustrative purposes where we hae used a *classical* analogy to shed light on what is a purely *quantum* effect.

C. Symmetry

As highlighted earlier, the circular symmetry of the annulus makes all observable characteristics independent of angular position. This is simply because any chosen angular position $\phi \to \phi'$ simply by means of a rotation. Alternatively, this can be understood by noting that any specified *numerical value* for an angular position is simply the result of an artificially imposed coordinate system, which can be reassigned to another position by rotating the axes. In concrete terms, the annulus has $SO(2)$ symmetry. The table below summarises the symmetry characteristics for the annulus.

$SO(2)$	E	$R(\phi)$	Basis
Γ^{+m}	1	$e^{im\phi}$	$(x+iy)^m$
Γ^{-m}	1	$e^{-im\phi}$	$(x-iy)^m$

TABLE II: Character table for the $SO(2)$ symmetry group of the circular annulus.

Since the irreducible representations for $SO(2)$ are 1D, there is yet again no normal degeneracy. However, note that since $E \propto m^2$, there are two values of m available for every non-zero value of E. Consequently our system has *accidental degeneracy*, where $m < 0$ corresponds to a particle whose angular position varies in an anticlockwise sense and $m > 0$ corresponds to a particle whose angular position varies in a clockwise sense.

IV. SPHERE

After working through the case of the circular annulus, it was inticing to explore how the *pseudo-potential* of the circular annulus would extend to a spherical shell which is obtained by rotating the circular annulus about an axis in the $x-y$ plane. Explicit solution of the TISE actually shows that the sphere has no effective *pseudo-potential*. This ofcourse, agrees with the prediction made by (B.16) in III-B:

$$U = \frac{1}{4}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)^2 = \frac{1}{4}\left(\frac{1}{a} - \frac{1}{a}\right)^2 = 0 \tag{.1}$$

where $R_1 = R_2 = a$ for the sphere.

A. Eigenfunctions

Keeping with tradition, we let the symmetry of the system dictate the choice of coordinates. Consequently, the most natural choice was to use spherical polar coordinates (r, θ, ϕ). The boundary conditions remain the same as those for the circular annulus. However, to obtain the full solution, we have to impose the condition that the wavefunctions are well defined at every point inside the spherical shell. This thus ensures that we have well behaved analytic solutions.

The solution for the TISE proceeds by writing the Laplace-Beltrami operator in spherical polars:

$$-\frac{\hbar^2}{2m_0}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right]=E\psi \tag{A.1}$$

By defining a separable wavefunction $\psi(r,\theta,\phi)=R(r)T(\theta)\Phi(\phi)$ and imposing the aforementioned boundary conditions, we obtain (see Appendix A-3),

$$\psi_{n,\ell,m}=C\Big[y_\ell(k_{\ell,n}a)j_\ell(k_{\ell,n}r)-j_\ell(k_{\ell,n}a)y_\ell(k_{\ell,n}r)\Big]P_\ell^m(\cos\theta)e^{im\phi} \tag{A.2}$$

where n, ℓ and m are the wavenumbers in the radial, polar and azimuthal directions respectively. C is a normalisation constant which again depends on the specific choice of wavenumbers. j_ℓ and y_ℓ are Spherical Bessel functions of order ℓ of the first and second kinds respectively, and P_ℓ^m are associated Legendre polynomials of order m and degree ℓ.

The figure below shows $|\psi_{n,\ell,m}|^2$ for a spherical shell of inner radius $a=1.0$ and outer radius $b=1.5$, with quantum numbers (n,ℓ,m): (a) (1,0,0), (b) (1,1,0), (c) (1,1,1), (d) (2,2,0), (e) (2,2,1) and (f) (2,2,2). We can immediately see that the polar ground state $(1,0,0)$ with $\ell=0$ has spherical symmetry. Increasing the radial wavenumber n, again results in an increase in the number of *coloured rings* as before. Varying the polar wavenumber m has the effect of simply rotating $|\psi_{n,\ell,m}|^2$ about the z axis. This is because, as we saw in the case of the circular annulus, $|\psi_{n,\ell,m}|^2$ is independent of ϕ. Finally, varying ℓ results in a variation in the number of *pockets*, i.e. regions with strongly localised probability density. These solutions of course are reminiscent of the same for the hydrogen atom.

For the energy eigenvalues in the thin shell limit, we obtain (see Appendix A-3),

$$E_{n,\ell}=\frac{\hbar^2k_{n,\ell}^2}{2m_0a^2}=\frac{\hbar^2}{2m_0}\left[\left(\frac{n\pi}{\delta a}\right)^2+\frac{\ell(\ell+1)}{a^2}\right]+O(\delta) \tag{A.3}$$

where $k_{n,\ell}$ is obtained in a manner similar to $k_{m,n}$ in the case of the circular annulus. We can immediately see that as predicted by our formula, $U=0$ for the spherical shell since the principle radii of curvature are equal and constant at every point. A physical interpretation for this can be obtained by realising that the spherical shell is obtained by wrapping around infinitely many of our linear strips on the $2-ball$. The race car driver now moving on this *track* can always find an effectively *straight* path to move along. This path of course is a *geodesic* on the surface of the sphere. Since the surface of the sphere is totally uniform, we can therefore find equivalent geodesics along every direction. In other words, the isotropy of the sphere results in the driver going around in infinite loops with effectively no track walls to crash into. Hence, the absence of a pseudo-potential.

B. Symmetry

The eigenstates for the spherical shell are spherically symmetric only in the polar ground state, i.e. when $\ell=0$. This is because we can obtain the spherical shell from the circular annulus only by rotating it about an axis *in the plane*, and not about the z axis. This therefore breaks the symmetry. Consequently, the spherical shell has $SO(3)$ symmetry with the basis functions given by the spherical harmonics $Y_{\ell,m}(\theta,\phi)$. The system now has normal degeneracy of order $(2\ell+1)$ due to the irreducible representations having dimensionality $(2\ell+1)$. The table below summarises these symmetry characteristics for the spherical shell [1].

$SO(3)$	E	$R(\varphi)$	Basis
Γ^ℓ	1	$\frac{\sin\left[\left(\ell+\frac{1}{2}\right)\varphi\right]}{\sin\left(\frac{1}{2}\varphi\right)}$	$Y_{\ell,m}(\theta,\phi)$

TABLE III: Character table for the $SO(3)$ symmetry group of the spherical shell.

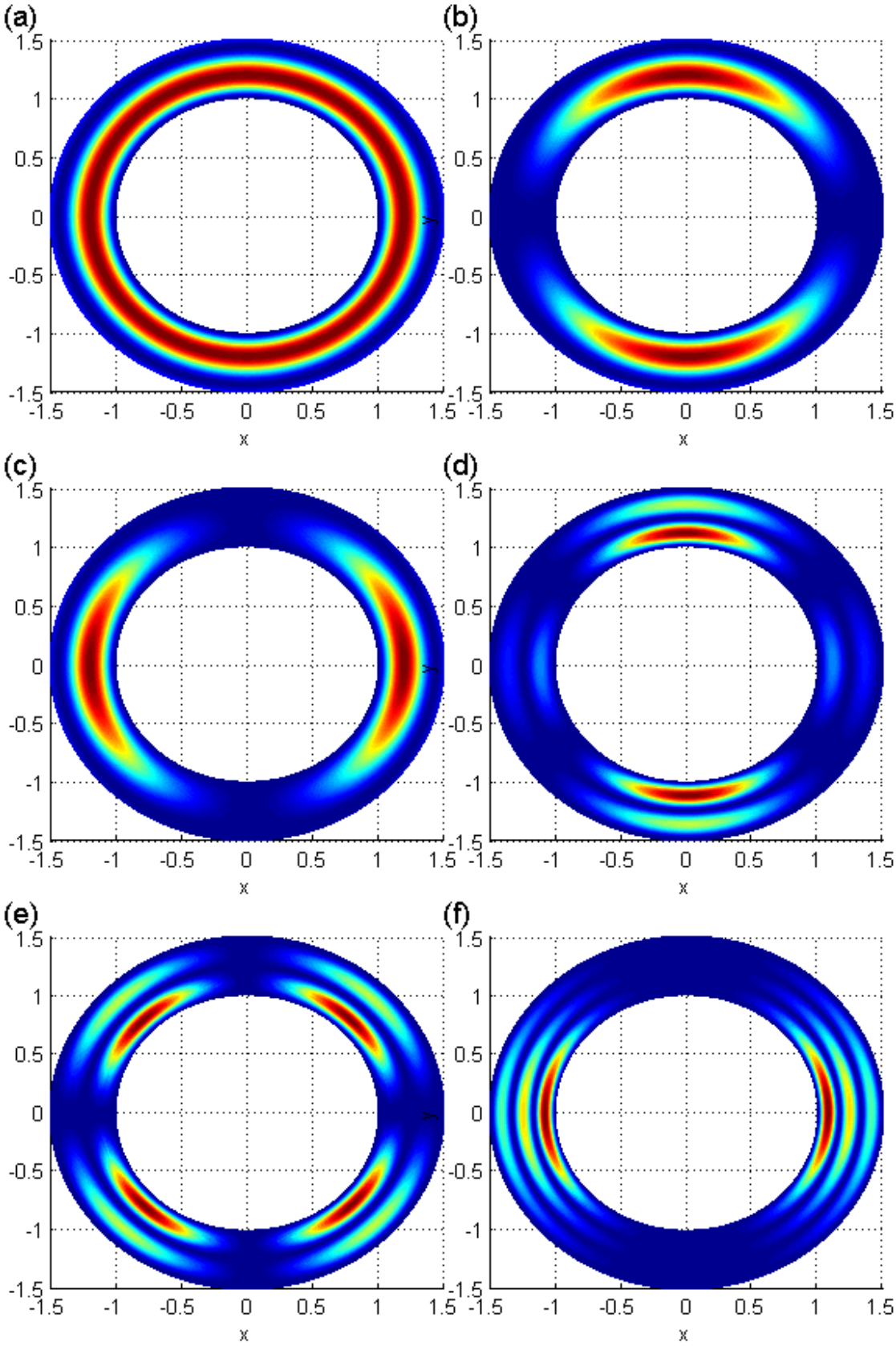

FIG. 4: Probability density functions $|\psi_{n,\ell,m}|^2$ for a spherical shell of inner radius $a = 1.0$ and outer radius $b = 1.5$, with quantum numbers (n, ℓ, m): (a) (1,0,0), (b) (1,1,0), (c) (1,1,1), (d) (2,2,0), (e) (2,2,1) and (f) (2,2,2). Note that these probability density functions are projections of the r and θ dependence with ϕ kept constant. This is because, as we saw in the case of the circular annulus, $|\psi_{n,\ell,m}|^2$ is independent of ϕ.

V. ELLIPTICAL ANNULUS

The elliptical annulus was interesting due to obvious reasons - the curvature is position dependent and can be controlled by varying the eccentricity. It was therefore key in bringing to light the full extent of the importance of curvature. Importantly, it could also be used to study the transition of the *pseudo-potential* from an elliptical annulus of zero eccentricity (a circular annulus) to those with more extreme eccentricities close to a value of 1.

The boundary conditions in this case were the same as before - the wavefunction was forced to vanish at the sidewalls and have no *branch-points*, i.e. $\psi(\phi) = \psi(\phi + 2\pi)$ at all points. Another important feature of our nanostructures throughout has been uniform thickness. This was trivial to ensure in the previous three cases. However, for the elliptical annulus, this required a special choice of coordinates as the more common choices, i.e. polar coordinates used for concentric ellipses and elliptical coordinate used for confocal ellipses, both result in annuli with varying thickness. This was a critical issue of course since we didn't want any effects due to the varying thickness of the annulus to cloud the effects due to variations in curvature.

A. Coordinate System

The coordinate system used for both the elliptical annulus and the *elliptic* helical annulus is described below [6]. The only difference is that the helix extends to 3D space due to the presence of a vertical pitch h in the z direction. In the case of the ellipse, the z coordinate remains constant throughout, and therefore doesn't enter into any calculations.

$$x = d\cosh\xi\cos\eta + \frac{\rho\sinh\xi\cos\eta}{\sqrt{\sinh^2\xi + \sin^2\eta}}$$

$$y = d\sinh\xi\sin\eta + \frac{\rho\cosh\xi\sin\eta}{\sqrt{\sinh^2\xi + \sin^2\eta}}$$

$$z = h\eta \tag{A.1}$$

The coordinate system above contains the following parameters:

- d - distance between the foci,
- ξ - measure of eccentricity $\epsilon = 1/\cosh\xi$,
- η - angular position,
- ρ - linear displacement for a fixed η,
- h - pitch of the helix

We can therefore use the above parameters to construct our desired elliptical annulus:

- set the distance between the foci as d,
- set the eccentricity ϵ by fixing $\xi = \cosh^{-1} 1/\epsilon$,
- set the thickness of the annulus by ensuring the inner and outer ellipses have linear displacement ρ and $\rho + \delta$ respectively

We can now move around from *side to side* by shifting between different values of ρ and *circle around* the annulus by varying the value of η. In the case of the helix, we can also vary h to move *up* the spiral.

B. Eigenfunctions

The complex nature of the coordinate system leads to an equally complicated TISE which also holds for the case of the helix,

$$\begin{aligned}
&\frac{\kappa\Big[\kappa\rho + d(\gamma+\sin^2\eta)^{3/2}\Big]}{\Big[d(\gamma+\sin^2\eta)^{3/2}+\kappa\rho\Big]^2 + h^2(\gamma+\sin^2\eta)^2}\partial_\rho\psi + \partial_\rho^2\psi \\
&+\frac{\sin 2\eta(\gamma+\sin^2\eta)\Big[d(\gamma+\sin^2\eta)^{3/2}+\kappa\rho\Big]\Big[\kappa\rho - d/2(\gamma+\sin^2\eta)^{3/2}\Big]}{\Big\{\Big[d(\gamma+\sin^2\eta)^{3/2}+\kappa\rho\Big]^2 + h^2(\gamma+\sin^2\eta)^2\Big\}^2}\partial_\eta\psi \\
&+\frac{(\gamma+\sin^2\eta)^2}{\Big[d(\gamma+\sin^2\eta)^{3/2}+\kappa\rho\Big]^2 + h^2(\gamma+\sin^2\eta)^2}\partial_\eta^2\psi + \frac{2mE}{\hbar^2}\psi = 0
\end{aligned} \tag{B.1}$$

The equation above is obtained through *brute force* and *endruance* and essentially contains no interesting mathematics. However, for the sake of completeness, we shall *outline* the method used:

- Notice that the essence of the TISE is the Laplace-Beltrami operator $\nabla^2 = \frac{1}{\sqrt{g}}\partial_i\sqrt{g}g^{ij}\partial_j$.
- Compute dx, dy and dz in terms of $d\rho$, $d\eta$ and dh.
- Use these expressions to compute the line element $ds^2 = g_{ij}dx^i dx^j$.
- Since x, y and z are mutually orthogonal, g_{ij} is a diagonal matrix. This fact can be easily used to collect together *like* terms in the expression for ds^2 and thereby compute the metric.
- Use this expression for the metric to compute its determinant g and the inverse metric g^{ij} whose entries are reciprocals of the entries in the metric matrix.
- Substitute the expressions for g and g^{ij} into the Laplace Beltrami operator and expand it.
- Hope for the best!

The solution for the elliptical annulus is obtained by setting $h = 0$ since there is no displacement along the z direction. The TISE now can not be solved analytically. It is therefore solved numerically by using a variational method [14],[15]:

Basis

Choose a complete (ideally orthonormal) basis set $\Phi = \{\phi_i\}$ such that for any ψ,

$$\psi = \sum_i a_i\phi_i \tag{B.2}$$

where $a_i \in \mathbb{C}$ are constants for a given ψ. This of course demands that the bases ϕ_i obey the boundary conditions if they are to sum up to a *square integrable* wavefunction.

Eigenvalue Problem

Using the basis set, solve the eigenvalue problem,

$$\mathbf{Ha} = E\mathbf{Sa} \tag{B.3}$$

where $\mathbf{H}$ is the Hamiltonian matrix, $\mathbf{a}$ is the N component vector comprising the coefficients a_i of the bases ϕ_i, E is the corresponding energy eigenvalue and $\mathbf{S}$ is a *normalisation* matrix.

Matrix Elements

Compute the matrix elements for $\mathbf{H}$ and $\mathbf{S}$,

$$H_{mn} = \int \phi_m^*\hat{H}\phi_n d^n r \tag{B.4}$$

$$S_{mn} = \int \phi_m^*\phi_n d^n r \tag{B.5}$$

where the integrals are over n dimensional space.

The eigenstates that satisfy the boundary conditions are [6],

$$\phi_{t,n} = \begin{Bmatrix} \cos t\eta \\ \sin t\eta \end{Bmatrix} \sin\left(\frac{n\pi(\rho+\rho_0)}{2\rho_0}\right) \tag{B.6}$$

where $n \in \mathbb{Z}^+$ and $t \in \mathbb{Z}_0^+$ are the radial and azimuthal wavenumbers respectively, and ρ_0 is the *radial* parameter for the ellipse which lies midway between the boundaries of the annulus such that $\phi(\eta, \rho_0) = \phi(\eta, -\rho_0) = 0$ where $\rho_0 = \delta/2$. Note that for every eigenstate $\phi_{t,n}$ there are two possible forms - one with $\cos t\eta$ and another with $\sin t\eta$, except for the $t = 0$ case when the $\sin t\eta$ term vanishes.

We can now use the basis states above to compute the matrix elements of **H** and **S** where it is important to note that the orthogonality of the states results in **S** being a diagonal matrix thereby further simplifying the solution of the eigenvalue problem.

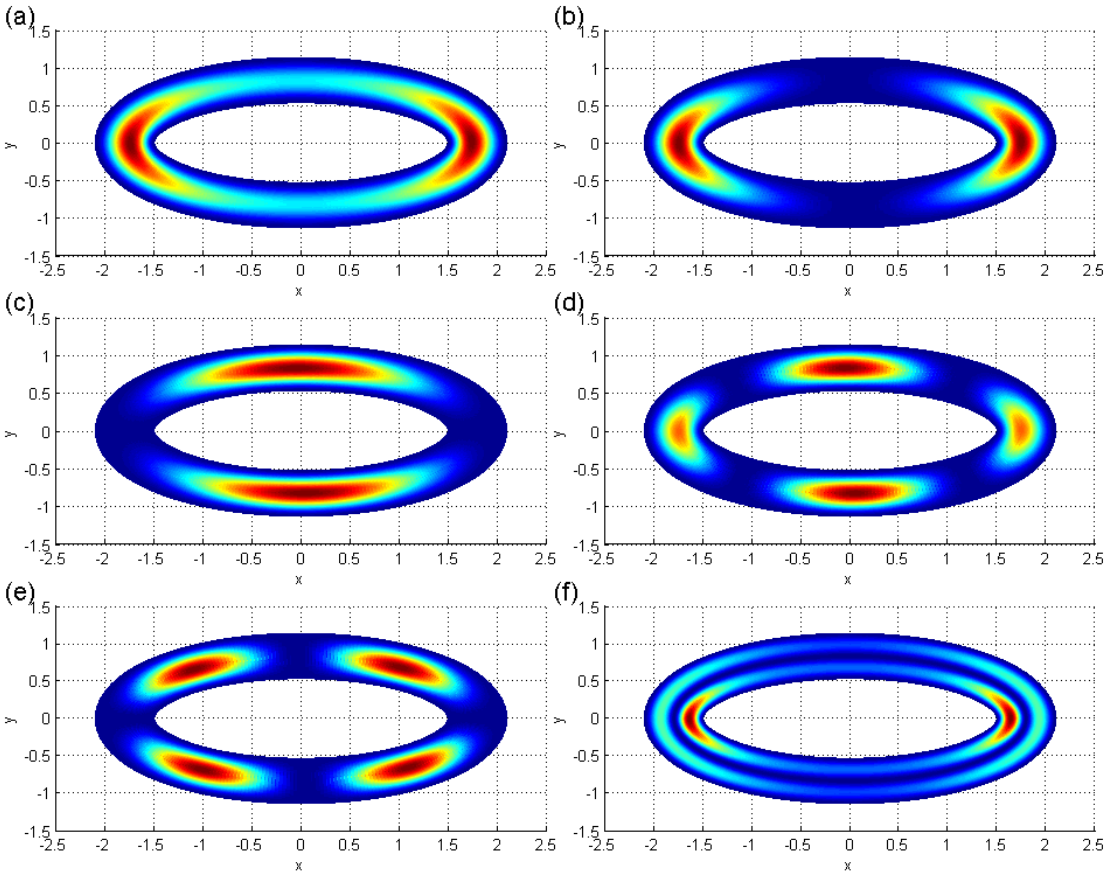

FIG. 5: Probability density functions $|\psi_{t,n}|^2$ for an elliptical annulus with $d = 1.59$, thickness $\delta = 0.6$ and eccentricity $\epsilon = 0.89$ with quantum numbers (t, n): (a) (0,1), (b) (1:cos η,1), (c) (1:sin η,1), (d) (2:cos 2η,1), (e) (2:sin 2η,1) and (f) (0,2). Note that (t:cos $t\eta$,n) means the *cosine* state with quantum numbers t and n, and likewise for (t:sin $t\eta$,n) which indicates the *sine* state with quantum numbers t and n.

The figure above shows $|\psi_{t,n}|^2$ for an elliptical annulus with parameters: $d = 1.59$, $\delta = 0.6$ and $\epsilon = 0.89$. Note for the ground state, how the significantly higher eccentricity leads to a severe *localisation* of the probability amplitude. Unlike the circle, the curvature of the ellipse varies with position and is highest near the *bends* closest to the focii. Consequently, we see the effects of a much higher *pseudo-potential* in this region leading to a significantly higher probability amplitude effectively implying the presence of a particle with a *localised presence*. In contrast to this, the regions furthest away from the foci, act like *straight tracks* similar to the linear strip where the probability amplitude is relatively much lower. Consequently, the elliptical annulus encompasses aspects of both the linear strip and the circular annulus.

As before, increasing the radial wavenumber n simply leads to an increase in the number of *rings*, and increasing the azimuthal wavenumber t results in an increase in the number of *localisation pockets* where for $t = 1$ we have 2 pockets and 4 in the case of $t = 2$. It is also interesting to note the effect that the $\sin t\eta$ and $\cos t\eta$ terms have on the localisation characteristics of $|\psi_{t,n}|^2$. In characteristic fashion, the *cosine* states peak at $\eta = 0$ and $\eta = \pi$, where as the *sine* states peak at $\eta = \pi/2$ and $\eta = 3\pi/2$ for $t = 1$. For $t = 2$, the situation becomes more complicated since the period is halved, which is why we also have a doubling in the number of *localisation pockets*. All these variations that can be achieved by simply varying the geometry of the nanostructure are of immense interest and have already found significant applications which we will discuss in due course.

C. Effect of Curvature on the Pseudo-potential

The results above provide some very valuable insight into the effects that curvature has on the probability amplitude and the particle's energy. We now know that there is a very strong correspondence between the curvature of the nanostructure and the value of the pseudo-potential, which consequently has a profound effect on the particle dynamics and probability amplitude. Figure 6 below perfectly demonstrates this where the ground-state probability density $|\psi|^2$ is shown for six different ellipses with different eccentricities ϵ: (a) 0.00, (b) 0.55, (c) 0.65, (d) 0.75, (e) 0.85 and (f) 0.89. We can clearly see how increasing the eccentricity increases the curvature at the points closest to the foci while reducing it at points much further away. This consequently results in much higher values for the *pseduo-potential* in the region with a high curvature where as the reduced curvature in the *straight track* region results in a further relaxation in the value of the confining pseudo-potential. Consequently we see that the probability density is significantly higher around the bends.

Although the solutions for the elliptical annulus were obtained numerically, it would be reasonable to assume that the formula for the energy would be similar to that for the circular annulus since an ellipse is topologically the same as a circle, but with a varying *radius* and thus a position dependent curvature. It is worth recalling the thin shell ($b/a - 1 = \delta \ll 1$) energy formula for the circular annulus,

$$E_{m,n} = \frac{\hbar^2 k_{m,n}^2}{2m_0} = \frac{\hbar^2}{2m_0}\left[\left(\frac{n\pi}{a\delta}\right)^2 + \left(\frac{m}{a}\right)^2 - \left(\frac{1}{2a}\right)^2\right] + O(\delta) \tag{C.1}$$

We can now use this to conjecture that the local energy for the ellipse would have a similar form such that,

$$E_{t,n} = \frac{\hbar^2}{2m_0}\left[\left(\frac{n\pi}{a(\rho,\eta)\delta}\right)^2 + \left(\frac{t}{a(\rho,\eta)}\right)^2 - \left(\frac{K(\rho,\eta)}{2}\right)^2\right] + O(\delta) \tag{C.2}$$

where the *local radius* a and the local curvature K are functions of ρ and η (position coordinates) for a given distance between the foci d and a given eccentricity ϵ. This relationship of course reiterates the strong correspondence between the pseudo-potential and the energy.

We shall now further scrutinise the effect of curvature by utilising the eccentricity of the ellipse as a variable parameter. We now fix our values for d and ρ, and investigate how varying the eccentricity ϵ results in variations in the pseudo-potential U at a given angular position η, and also how U varies with η for an ellipse with a given eccentricity ϵ. Figure 7 below summarises these results. This therefore reiterates our previous findings and further brings to light the significance of curvature in determining the particle dynamics. Thus higher curvature results in more extreme characteristics with higher energy *plateaus* in the *straight track* region and lower energy *valleys* near the *bends*. This in fact also allows explains the symmetry breaking in the ellipse due to the splitting of the energy of the *sine* and *cosine* states as shown in Figure 8. For the cosine states with *localisation pockets* nearer to the bends, the energy decreases with increasing eccentricity since the weighting of the pseudo-potential increases with the increase in eccentricity. However, for the *sine* states, which have *localisation pockets* in the *straight track* region, the exact opposite occurs and the energy increases instead.

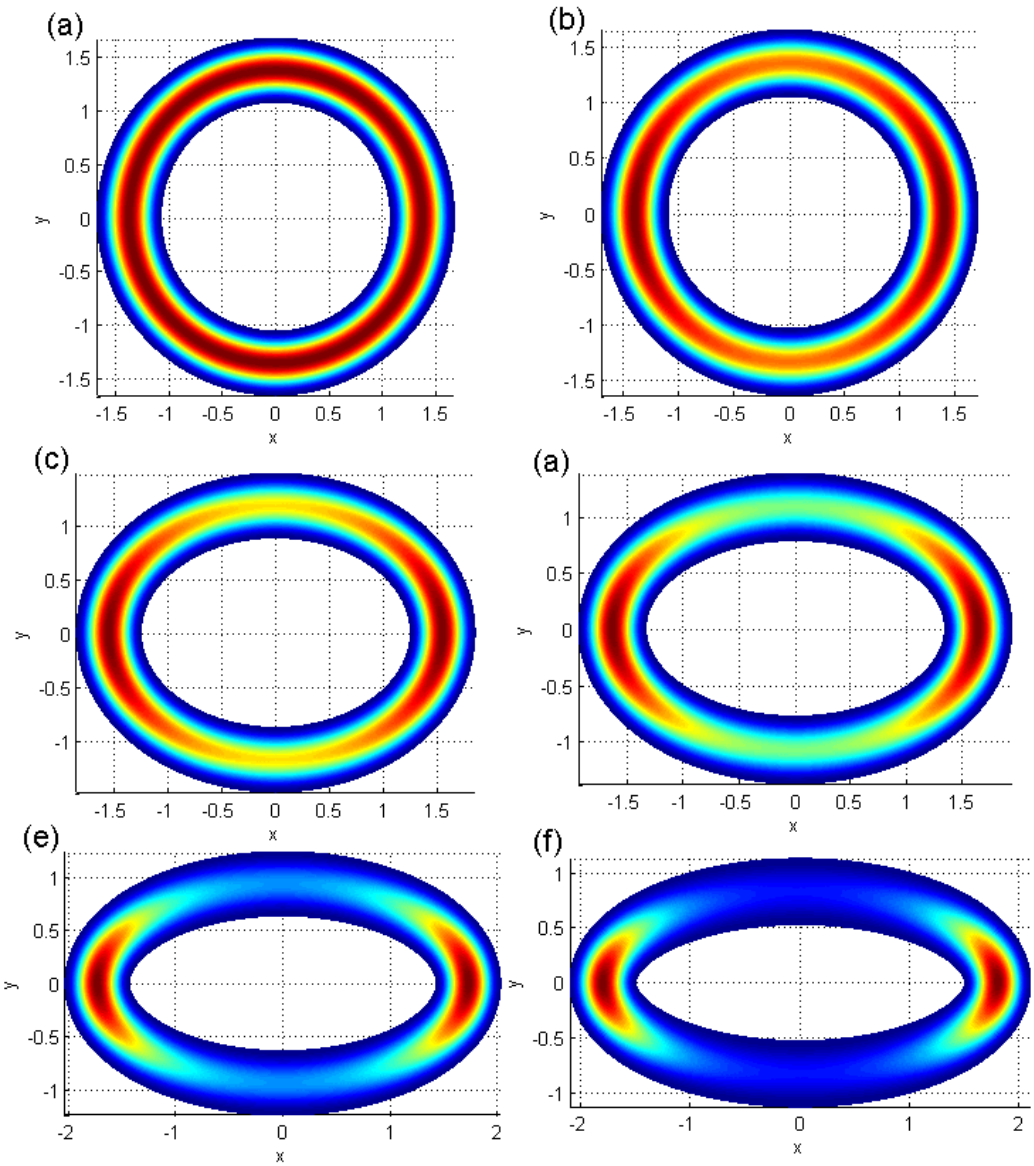

FIG. 6: Probability density functions $|\psi|^2$ for ellipses with $d = 1.59$, $\delta = 0.3$ and varying eccentricity ϵ: (a) 0.00, (b) 0.55, (c) 0.65, (d) 0.75, (e) 0.85 and (f) 0.89.

D. Symmetry

The symmetry group of the ellipse is $C_{2v} \cong D_2$. Therefore, the ellipse has the exact same symmetry characteristics as the rectangular strip with no normal degeneracy and only accidental degeneracy. Figure 8 also demonstrates a *spontaneous symmetry breaking* when the eccentricity moves away from $\epsilon = 0$ since there is an abrupt transition from the $SO(2)$ symmetry of the circle to the C_{2v} symmetry of the ellipse. This is because the angular momentum operator which is compatible with the Hamiltonian for the annulus with circular symmetry, no longer shares the same eigenstates when this symmetry is broken. Consequently, this results in a splitting in the energy of the *sine* and *cosine* states.

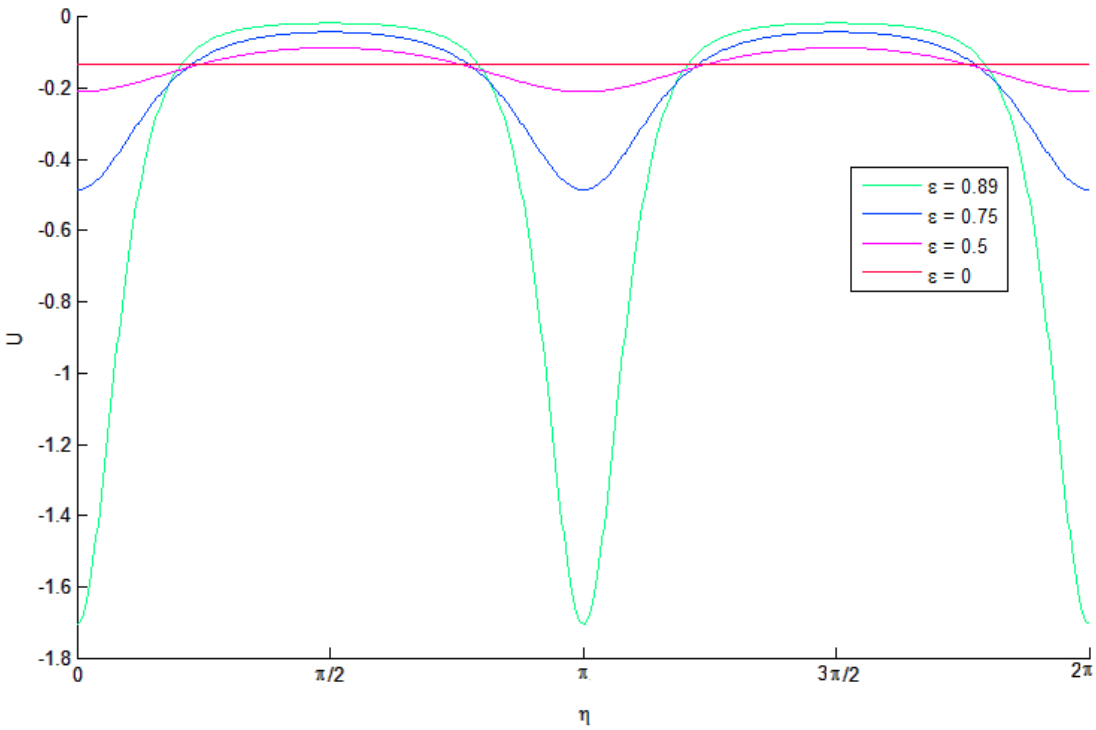

FIG. 7: Plot showing variation in pseudopotential U with angular position η for ellipses with different eccentricities $\epsilon = 0.00$, 0.50, 0.75 and 0.89. Focusing on the variation in U with varying ϵ at a given η reiterates the results in Figure 6. We can also see how drastically the potential changes near the *bends* at $\eta = 0$ and $\eta = \pi$. The circle with eccentricity $\epsilon = 0.00$ provides a base reference.

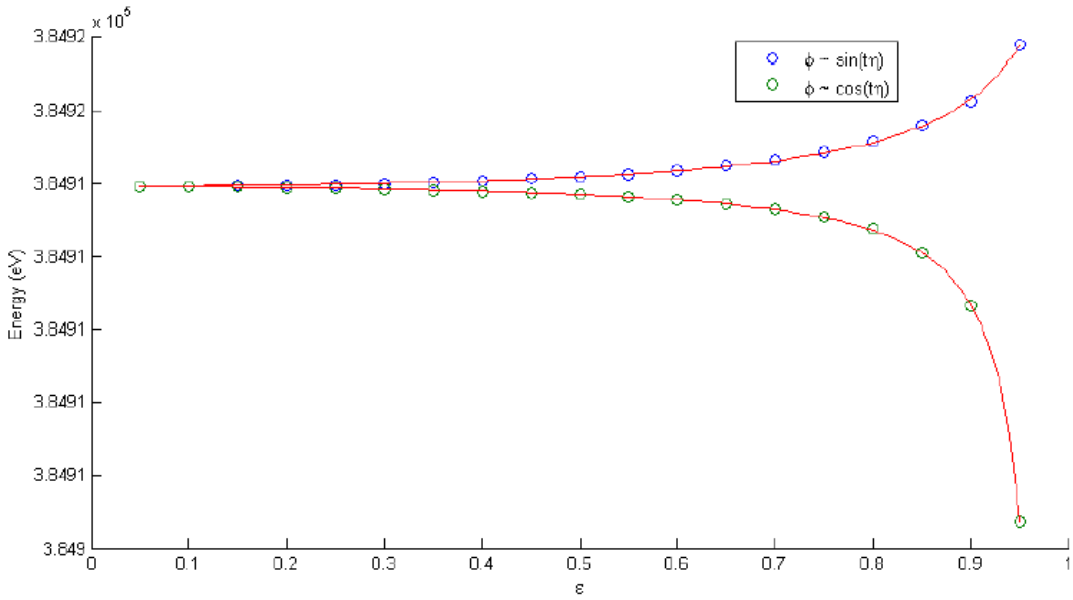

FIG. 8: Variation of energy with eccentricity for the $\phi_{1,1}$ state for an ellipse with $\delta = 0.01$ and $\epsilon = 0.75$. The energy splitting of the degenerate *sine* and *cosine* states is clearly visible.

VI. HELICAL ANNULUS

It is interesting to note that the extreme localisation of the probability amplitude in the ellipse can be used for assembling *quantum dots* which are essentially the physical realisation of our hypothesised *severely localised* particles. Such quantum dots have applications in several areas of Solid State research and also form a significant part of active research in Condensed Matter theory. Often an important challenge is to align quantum dot arrays in a stable structure which can be used for various purposes including the production of low cost and high power lasers. This thus led us to the (elliptic) helical annulus. We realised that if by using ellipses with strong eccentricities we could assemble quantum dots, we could also use a helical annulus with elliptic spirals to possibly produce a vertical array of quantum dots by exploiting the localisation characteristics.

The coordinate system for the helical annulus is the same as that described in (A.1) of V-A. The pitch is now non-zero of course. The extension into 3D requires a slight refinement of the boundary conditions. We still demand our wavefunctions to vanish at the sidewalls and have the same value at the top of the spiral as it is at the bottom, thereby maintaining a continuity in the wavefunction in line with the Born-von Karman boundary conditions. However, since the spiral now has s turns, we must alter the angular boundary condition such that $\psi(\rho,\eta) = \psi(\rho,\eta+2\pi s)$ now. This presence of twists consequently also requires us to slightly tweak the basis functions $\phi_{t,n}$ such that,

$$\phi_{tn} = \begin{Bmatrix} \cos(t\eta/s) \\ \sin(t\eta/s) \end{Bmatrix} \sin\left(\frac{n\pi(\rho+\rho_0)}{2\rho_0}\right) \tag{.1}$$

A. Eigenfunctions

The TISE then is the same as that mentioned in the previous section with the solution proceeding in a similar fashion too. Figure 9 below shows a selection of probability density functions.

The probability density functions $|\psi_{t,n}|^2$ shown are for an (elliptic) helical annulus with focal distance $d = 1.59$, thickness $\delta = 0.6$, eccentricity $\epsilon = 0.75$ and pitch $h = 0.1$ with quantum numbers (t,n): (a) (0,1), (b) (1:cos η,1), (c) (1:cos η,2) and (d) (1:sin η,2). Note that the notation used in labeling the states is the same as in the case of the ellipse. Note also that (a) and (b) are helices with 2 spirals and (c) and (d) are helices with 3 spirals instead. These probability density functions infact are very much the same as those for an ellipse with the same quantum numbers except that they can now be visualised by considering one such ellipse, cutting it with scissors and then twisting the ends into a right handed spiral. Doing so would exactly reproduce what we have here.

B. Unification Scheme

Our progress through the nanostructures has followed a natural route so far with greater complexity at every step and each subsequent nanostructure containing features of the one before it along with added new features due to a richer structure in terms of curvature. It is therefore natural that all the nanostructures considered so far (except for the spherical shell which was an interesting detour) can effectively be recovered as degeneracies of the *elliptic* helical annulus. This is simple to understand: a helix with zero pitch would simply degenerate into an ellipse, and an ellipse with zero eccentricity would trivially result in a circle. However, there is no obvious way of transforming an elliptical or circular annulus into a linear strip since the strip is topologically different from the two. However, we can recover the linear strip from the helical annulus by *stretching* it along the z direction such that in the limiting case the twists become negligible, or we can simply untwist the helix to trivially acquire a linear strip! Note, however, that this strip would have slightly different boundary conditions since instead of vanishing at both ends along the z direction, the wavefunctions inside it must now have continuity across the ends. Figure 10 illustrates this transformation for the ground state of the 2-twist helical annulus with a pitch that is 10 times larger. Already, we can see the drastic effect that this has on causing a disintegration of the localisation of the probability density. The helix has *effectively* transformed into a linear strip

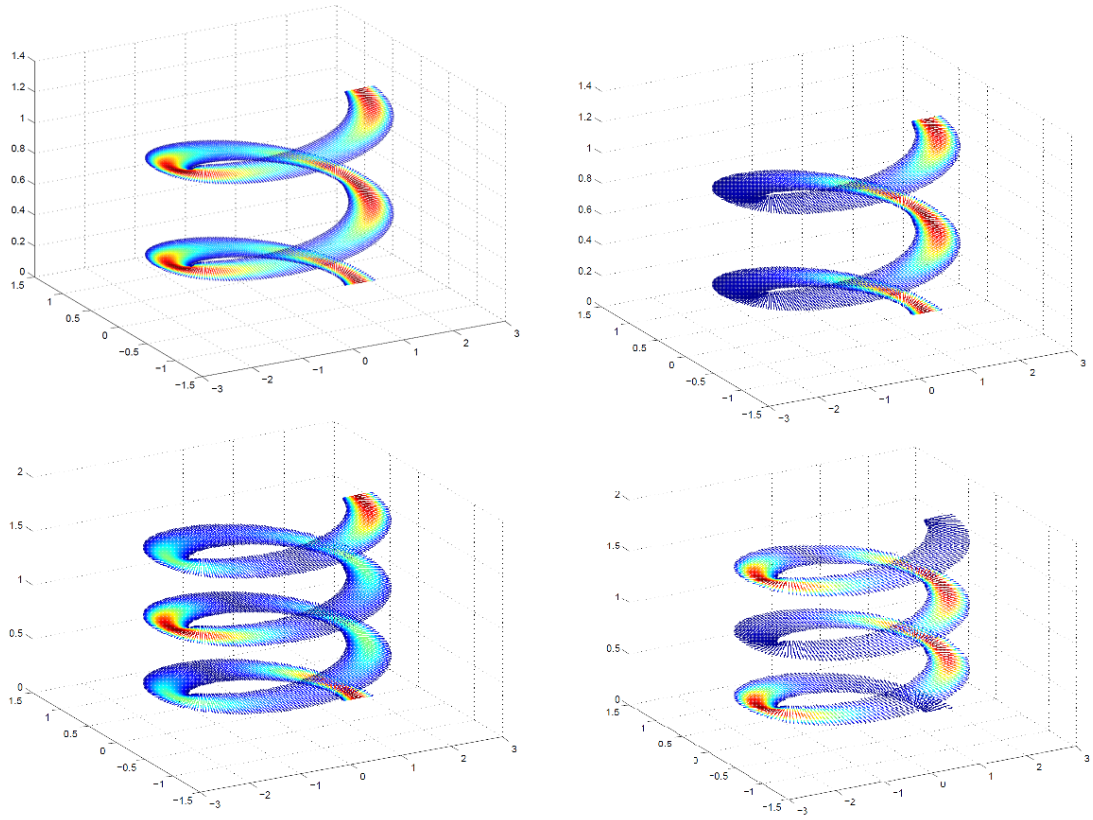

FIG. 9: Probability density functions $|\psi_{t,n}|^2$ for an (elliptic) helical annulus with focal distance $d = 1.59$, thickness $\delta = 0.6$, eccentricity $\epsilon = 0.75$ and pitch $h = 0.1$ with quantum numbers (t, n): (a) (0,1), (b) (1:cos η,1), (c) (1:cos η,2) and (d) (1:sin η,2).

with continuous ends with roughly uniform probability density throughout as would be expected in linear strip with end continuity along the z direction.

Figure 11 demonstrates the effect that pitch has on the energy of the states. It is interesting to see how all the states seem to converge to a single value of energy equal to the zero point energy of the ground-state as the pitch approaches very high values. This is a testament to our claims earlier. Also notice how all the *cosine* states hit a peak and then descend, where as all the *sine* states are monotonically decreasing. This is because for the *cosine* states, increasing the pitch of the helix results in much gentler curvature since the same number of twists now occur over a longer distance. This consequently reduces the *pseudo-potential* thereby increasing the energy. However, another effect of increasing the pitch is a decrease in the energy of angular motion. As this effect takes over from the energy increase effect, the curve commences its asymptotic descend towards the zero-point energy. For the *sine* states, however, since the *localisation pockets* are at the ends of the minor foci where the curvature is lowest, the effect of the increase in energy due to a decrease in curvature is throughout smaller than the decrease in energy due to the loss in azimuthal motion. Consequently since the loss of energy due to the loss in azimuthal motion wins at all times, we see monotonically decreasing *sine* states.

We therefore see that all the nanostructures (except for the spherical shell) can be obtained by transforming the elliptic helical annulus. These nanonstructures are effectively degenerate structural forms of the helical annulus.

C. Symmetry

The symmetry operations for the helix are more diverse since the helix extends out of the plane. Consequently, these are a combination of

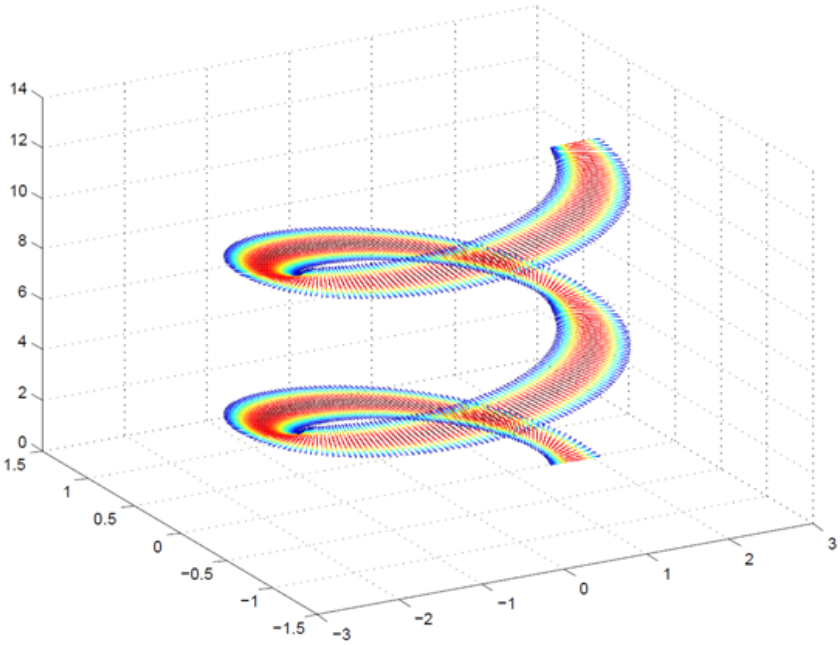

FIG. 10: Probability density function $|\psi_{t,n}|^2$ for an (elliptic) helical annulus with focal distance $d = 1.59$, thickness $\delta = 0.6$, eccentricity $\epsilon = 0.75$ and pitch $h = 1.0$ with quantum numbers $(t, n) = (0, 1)$

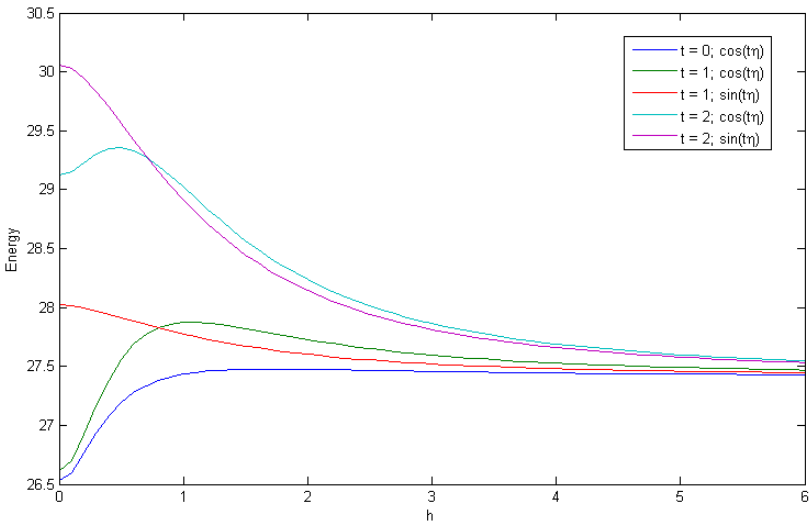

FIG. 11: Variation in energy with pitch for an elliptic helical annulus with $d = 1.59$, $\delta = 0.6$ and $\epsilon = 0.89$ with all states sharing the same radial quantum number $n = 1$.

VII. APPLICATIONS

The main application of these nanostructures, particularly the elliptical annulus and the elliptic helical annulus is in the assembly of quantum dots and quantum dot arrays. These quantum dots can then be used for a wide variety of purposes. We outline a handful of such applications below that have taken shape over the past few years and offer extremely exciting prospects in this line of research.

A. Quantum Dot Memories

Quantum Dot Memomries (QDMs) have recently emerged as an extremely viable alternative to the current methods of memory storage including Dynamic Random Access Memories (DRAMs), Solid State Drives (SSDs) and Flash memories. The main obstacle in their large-scale production is the difficulty in producing stable quantum dot (QD) arrays. However, with the recent advancements in the production of nanostructures akin to those mentioned above that can be used to assemble stable quantum dot arrays, the production of QDMs seems extremely viable. The figure below [23] is a schematic of the reading and writing processes involved in the QDMs.

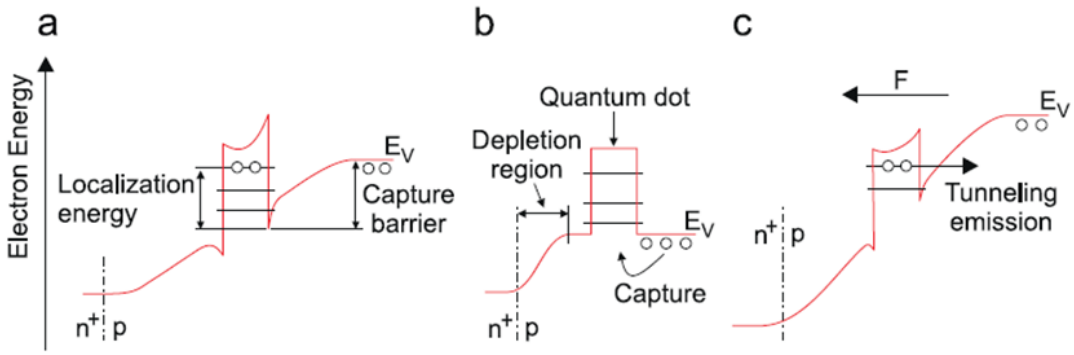

FIG. 12: Schematic illustration of the (a) storage, (b) write and (c) erase processes in the proposed QD based Flash-memory based on hole storage.

The memories are fabricated using a $p-n$ junction assembly in which the quantum dots are affixed on the p side of the depletion region with a zero bias voltage. The high potential region formed by the quantum dots prevents a flux of holes in this region of the nanostructure - see Figure 12(a). Consequently, holes trapped inside the quantum dot remain trapped inside, whilst those on the exterior find it difficult to break through the potential barrier as well. This constancy in the configuration of charge carriers ensures that the memories are resilient and have a long life time. In fact, the main reason behind the placement of the quantum dots on the p side is to ensure robustness since the higher effective mass of holes reduces the likelihood of tunneling through the quantum dot potential barrier, thereby ensuring that the integrity of the data can be maintained for long periods of time. Also, since the memory is able to retain its state without the use of power in a manner similar to Flash drives, and unlike DRAMs doesn't need to be *refreshed* every so often it also makes it extremely energy efficient.

The writing process in Figure 12(b) involves the application of a forward bias voltage that shrinks the depletion region thereby expelling the quantum dot from the depletion region. The lowering of the potential barrier due to the expulsion of the quantum dot from the depletion region thereby makes it easier for the quantum dot to capture holes for memory storage. Once the storage process is complete, the system is returned to the initial bias voltage which results in a re-encapsulating of the quantum dot into the depletion region, putting the memory back into storage mode.

Finally, erasing data - Figure 12(c) requires the application of a reverse bias while maintaining the quantum dot inside the depletion region. The reverse bias field lowers the effective potential barrier of the quantum dot thereby allowing holes to tunnel through the quantum dot more easily allowing the memory to be erased.

Therefore, this combination of storage, write and erase procedures in QDMs makes them superior to currently available methods of memory storage. The integrity of the data can be maintained for periods in excess of 10^6 years without the constant use of power in a manner similar to Flash drives, and unlike DRAMs, QDMs don't need to be *refreshed* every so often making them extremely energy efficient. Additionally, the access time for the data stored on QDMs is of the order of 10ns, which is less than half of that required by traditional DRAMs. Consequently, not only would such QDMs be more robust and energy efficient, they would also offer a faster alternative to accessing large amounts of data.

B. ZnO Nanostructures

Zinc Oxide and similar ionic compounds can be used to manufacture nano-belts [16], [17]. In the case of ZnO, the mechanical stress on the extremely thin material, $\sim$ 10nm, causes a charge separation resulting in polarised surfaces with a corresponding electric dipole moment. These nano-belts tend to roll into rings, spirals and helices. This is because the resulting geometry has a much lower electrostatic energy compared to that of a straight belt. In fact, the mechanical stress and the accompanying increase in elastic potential energy resulting from the transformation into more intricate geometries, is exceeded by the corresponding decrease in electrostatic potential energy thereby making the formation of rings, spirals and helices energetically favourable. In fact, it is this intricate energy balance that determines the exact details of the nanostructures that evolve from these nano-belts.

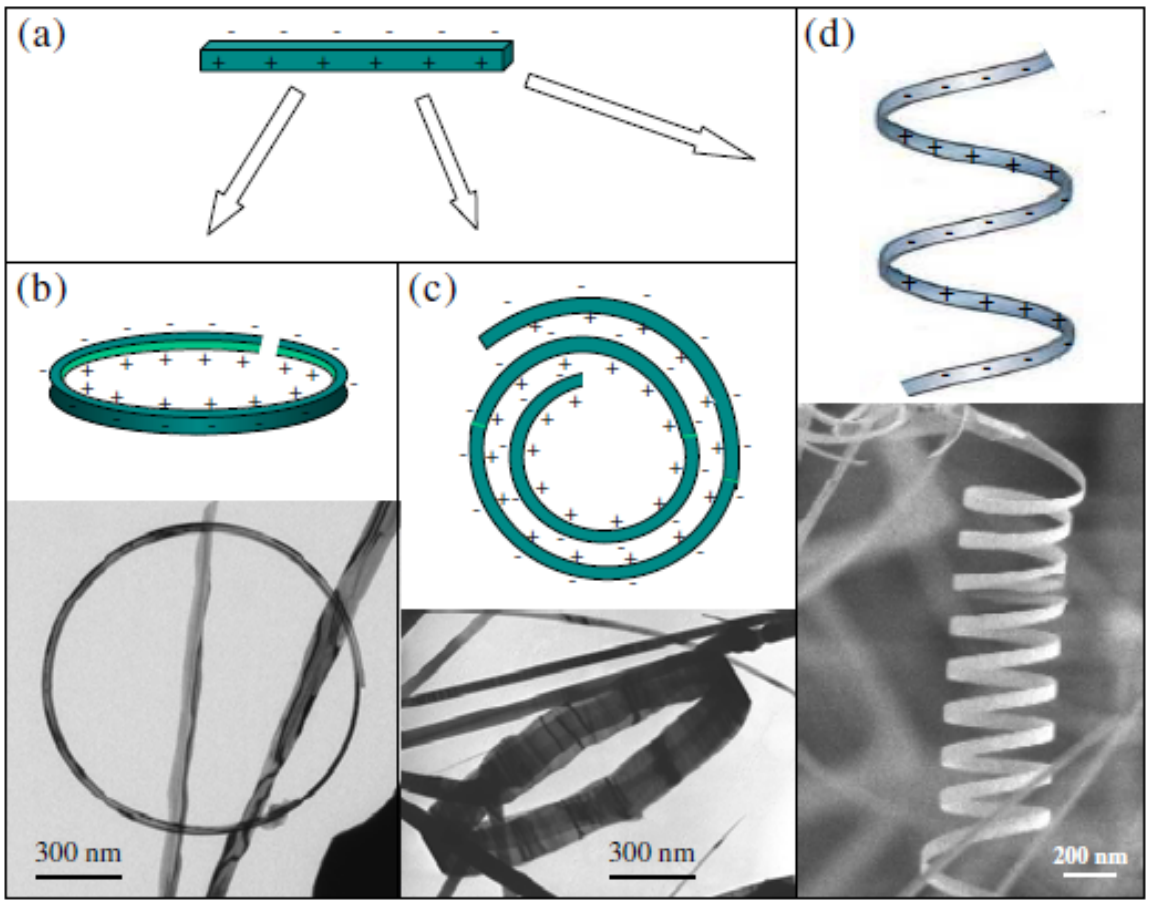

FIG. 13: (a) The model of a polar nano-belt. Polar surface induced formation of (b) a nano-ring, (c) a nanospiral and (d) nanohelixes of ZnO and their formation processes.

The resulting ZnO nanostructures have a direct band gap of about 3.37eV under standard conditions which can be utilised for emission and absorption at the *blue* end of the electromagnetic spectrum. This therefore makes these nano-belts and the derived nanostructures viable materials for use in photoluminescence in low energy and high resolution *flexible* displays and also in the production of absorption panels for solar cells and other electronic devices.

C. Quantum Dot Displays

Another possible use of quantum dot arrays is in the production of Quantum Dot Displays (QDDs). These comprise of what are essentially Quantum Dot Light Emitting Diodes (QD-LEDs). The quantum dots that occupy the band gaps in semiconducting materials are able to trap charge carriers in between [18]. However, only a limited number of carriers can be trapped inside these quantum dots owing to restrictions imposed by the finite confining potential, the size of the quantum dots, their shape and chemical composition [19].

Figure 14 below [22] is a schematic illustration of the workings of quantum dots and how they can be utilised for light emission in different parts of the spectrum by altering their size. Excitation of electrons in the semiconducting material elevates their energy to that of the lowest unoccupied state in the conduction band. However, these energetic electrons get trapped by the quantum dots instead. The confined electron is now closer to the hole in the valence band than it would otherwise have been had it escaped to the conduction band. Consequently, there is still an effective Coulomb attraction present between the two. This electron-hole pair confined within the quantum dot is called an exciton and emits electromagnetic energy equal to the energy difference of the electron-hole pair when the electron recombines with the hole after descending back to the valence band [18], [20]. As illustrated, the size of the quantum dot and the corresponding band gap are inversely related. We can therefore use this to our advantage for producing light of varying frequencies simply by varying the size of the quantum dot.

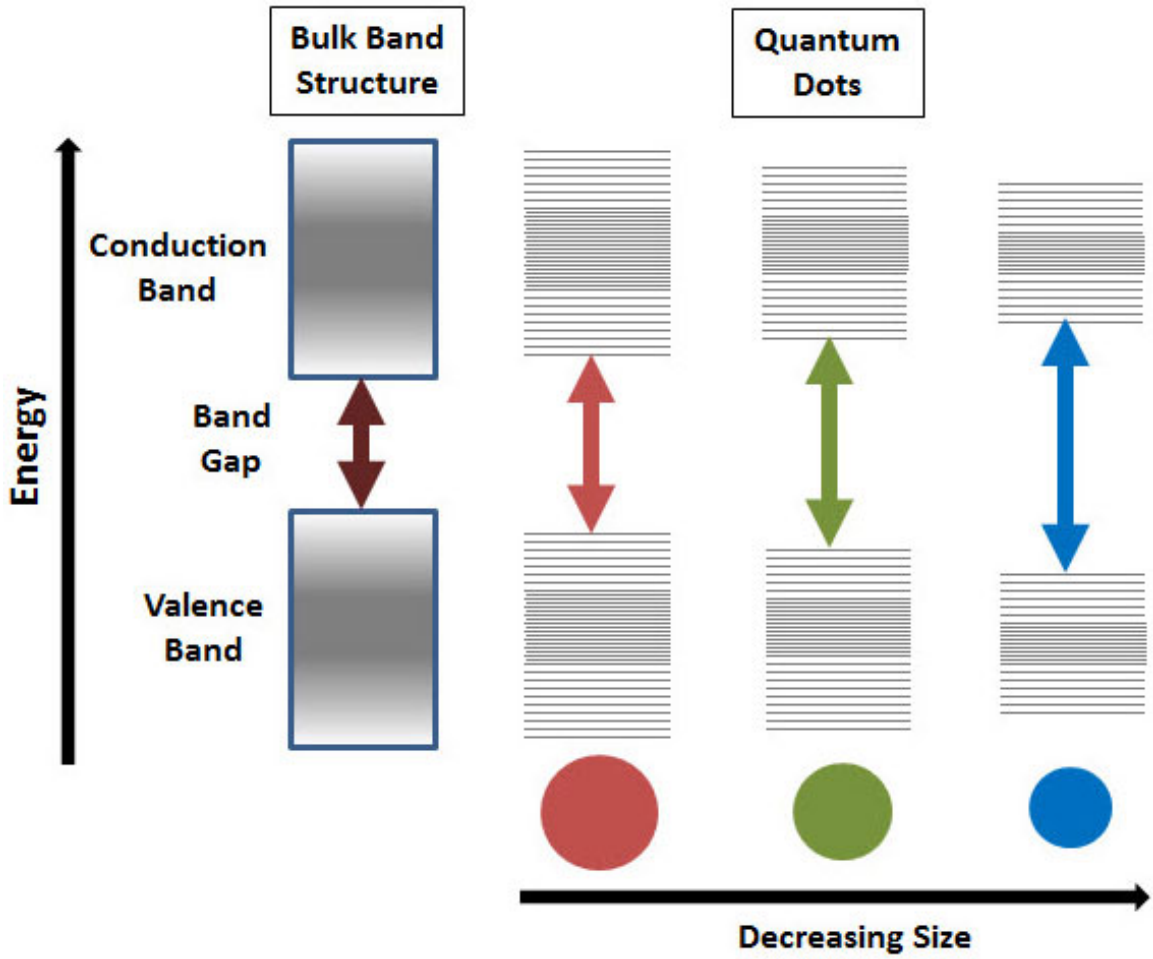

FIG. 14: Splitting of energy levels in quantum dots due to the quantum confinement effect. The semiconductor band gap increases with decrease in size of the nanocrystal.

Decreasing the size of the quantum dot effectively *squeezes* the energy level structure thereby exaggerating the confinement effect in the quantum dot. Although the reduced size leads to an

increase in the coulomb interaction, $\sim 1/d$, where d indicates the linear size of the quantum dot, the corresponding increase in confinement energy $\sim 1/d^2$ due to a reduction in the linear size dominates and consequently results in a widening of the band gap [21]. Consequently, by manipulating the size of the quantum dots we can achieve the desired band gap and thus the required frequency of light. It is important to note that these size variations can be fine-tuned very precisely to produce very specific colours thereby allowing the production of displays with a very high resolution. Since the resulting frequency spread is low, the images produced can be made very sharp. This also has the advantage of allowing the production of displays that are brighter and more energy efficient than current LED displays [18]. Perhaps the most exciting aspect of the use quantum dots in making displays is that they can be produced as solutions that can be adsorbed onto any flexible surface including plastics and glass thereby making the achievement of working flexible displays very much possible.

This length scale is of orders of 10 nm. Thus by varying the size of the quantum dots it is possible to control the wavelength of the emitted photons, ranging from red to blue in the visible spectrum.

Other advantages of Quantum dot displays is that the emission spectra is a very narrow Gaussian distribution meaning that the range of the wavelengths of photons emitted from a particular quantum size is not broad and hence almost definite colors can be obtained. Moreover the QD displays have low temperature dependence and higher efficiency than regular LED's [18]. They are also photostable, compatible with solution processing and can be deposited on any substrate including glass and plastic.

VIII. CONCLUSION

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APPENDIX A: DETAILED SOLUTIONS FOR THE NANOSTRUCTURES

1. Linear Strip

As mentioned in II-A we proceed by assuming a separable solution $\psi(x,y) = X(x)Y(y)$. Substituting this into the TISE yields,

$$\frac{-\hbar^2}{2m_0}\left(Y\frac{d^2X}{dx^2} + X\frac{d^2Y}{dy^2}\right) = EXY \tag{1.1}$$

Rearranging and collecting like terms, we obtain,

$$-\frac{X''}{X} = \frac{Y''}{Y} + \frac{2m_0E}{\hbar^2} \tag{1.2}$$

Since LHS = LHS(x) and the RHS = RHS(y), we can argue that for both the LHS and RHS to agree for all values of x and y, they must in fact equal a constant S which is our *separation constant.*The constant S can be negative, zero or positive. When we evaluate the wavefunctions for negative S, the resulting solution does not satisfy the boundary conditions. When S is zero, we obtain a trivial solution. However, using $S = k^2 > 0$, we obtain a non-trivial wavefunction that satisfies the boundary conditions. The solution for this progresses as outlined,

$$-\frac{d^2X}{dx^2} = k^2X \tag{1.3}$$

which can be solved by using a trial function $X = Ae^{px}$. This then simplifies to a linear combination of the form,

$$X(x) = A\sin kx + B\cos kx \tag{1.4}$$

where $A, B \in \mathbb{C}$. Enforcing the boundary conditions,

$$X(0) = B = 0$$

$$X(a) = A\sin ka + B\cos ka = A\sin ka = 0 \tag{1.5}$$

The boundary conditions are satisfied if $ka = m\pi$, $\forall m \in \mathbb{Z}$. Consequently we obtain the final form of our solution,

$$X_m(x) = C_1\sin\left(\frac{m\pi x}{a}\right) \tag{1.6}$$

Note that the trivial solution is included in our solution set where using $m = 0$ yields the trivial solution. Negative values of m simply mirror the positive values, and as a consequence of symmetry have no unique effect on the probability density. We can therefore reduce our solution set for m to $m \in \mathbb{Z}_0^+$.

Solving for Y, we follow on from (1.2),

$$\frac{d^2Y}{dy^2} + \left(\frac{2m_0E}{\hbar^2} - k^2\right)Y = 0 \tag{1.7}$$

Solving this in a similar manner to the equation for X, and applying the boundary conditions we obtain,

$$Y_n(y) = C_2\sin\left(\frac{n\pi y}{b}\right) \tag{1.8}$$

where $n \in \mathbb{Z}_0^+$. The complete solution is then,

$$\psi_{m,n}(x,y) = C\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) \tag{1.9}$$

The normalisation condition $\int_0^a \int_0^b |\psi|^2 dxdy = 1$ then yields,

$$C = \sqrt{\frac{4}{ab}} \tag{1.10}$$

which results in the total wavefunction,

$$\psi_{m,n}(x,y) = \sqrt{\frac{4}{ab}} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \tag{1.11}$$

2. Circular Annulus

As mentioned in III-A we proceed by assuming a separable solution $\psi(r,\phi) = R(r)\Phi(\phi)$. Substituting this into the TISE yields,

$$-\frac{\hbar^2}{2m_0}\left(\Phi\frac{d^2R}{dr^2} + \frac{1}{r}\Phi\frac{dR}{dr} + \frac{1}{r^2}R\frac{d^2\Phi}{d\phi^2}\right) = ER\Phi \tag{2.1}$$

Rearranging and collecting like terms, we obtain,

$$-\frac{\Phi''}{\Phi} = r^2\frac{R''}{R} + r\frac{R'}{R} + \frac{2m_0E}{\hbar^2}r^2 \tag{2.2}$$

Since LHS = LHS(r) and the RHS = RHS(ϕ), we can argue that for both the LHS and RHS to agree for all values of r and ϕ, they must in fact equal a constant S which is our *separation constant*.

Solving for Φ, we obtain,

$$\frac{d^2\Phi}{d\phi^2} = -S\Phi \tag{2.3}$$

where identical to the case for the linear strip $S = m^2 > 0$. The solution for Φ is then,

$$\Phi(\phi) = Ae^{im\phi} + Be^{-im\phi} \tag{2.4}$$

Enforcing the boundary condition $\psi(r,\phi) = \psi(r,\phi+2\pi)$, we are able to restrict our solution set for m to $\mathbb{Z}$. This therefore makes the $Be^{-im\phi}$ term in the equation above redundant. Consequently our solution simplifies to,

$$\Phi(\phi) = Ae^{im\phi}, \quad \forall m \in \mathbb{Z} \tag{2.5}$$

Solving now for R, we obtain,

$$r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} + (k^2r^2 - m^2)R = 0 \tag{2.6}$$

where $k = \sqrt{2m_0E}/\hbar$. The equation above is Bessel's equation with the general solution [4],

$$R(r) = CJ_m(kr) + DY_m(kr), \tag{2.7}$$

where C and D are constants of integration and J_m and Y_m are Bessel functions of order m of the first and second kind respectively. Although Y_m is singular at the origin, we can retain this term since the origin is not included in the region defined by our radial boundary condition $\psi(a,\phi) = \psi(b,\phi) = 0$ where $b > a > 0$. Consequently we obtain a pair of coupled equations for C and D which can be summarised as a single matrix equation,

$$\begin{pmatrix} J_m(ka) & Y_m(ka) \\ J_m(kb) & Y_m(kb) \end{pmatrix}\begin{pmatrix} C \\ D \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{2.8}$$

Therefore, for non-trivial C and D, the determinant of our coefficient matrix must vanish. This then leads to the condition,

$$J_m(ka)Y_m(kb) - J_m(kb)Y_m(ka) = 0 \tag{2.9}$$

where for a given m, the nth root of the equation gives us the eigenvalue $k_{m,n}$

To further simplify our solution for the radial part of the wavefunction, we note that C and D are related as follows,

$$D = -C\frac{J_m(ka)}{Y_m(ka)}, \tag{2.10}$$

where $J_m(k_{m,n}a)$ and $Y_m(k_{m,n}a)$ are constant for a given m and n. This therefore yields,

$$R(r) = \frac{C}{Y_m(ka)}\Big[Y_m(ka)J_m(kr) - J_m(ka)Y_m(kr)\Big] \tag{2.11}$$

Combining our solutions for R and Φ,

$$\psi_{m,n}(r,\phi) = R(r)\Phi(\phi) = Ce^{im\phi}\Big[Y_m(ka)J_m(kr) - J_m(ka)Y_m(kr)\Big] \tag{2.12}$$

where C is the normalisation constant.

In the thin shell limit, i.e. $b/a - 1 = \delta$, the roots of

$$J_m(z)Y_m(\lambda z) - J_m(\lambda z)Y_m(z) = 0 \tag{2.13}$$

are real and simple if m is real and $\lambda = b/a = 1+\delta$ is positive [4]. We therefore obtain the following asymptotic expansion for the nth root $z_{m,n}$ [4],

$$z_{m,n} = \frac{n\pi}{\lambda - 1} + \frac{4m^2 - 1}{8\lambda}\frac{\lambda - 1}{n\pi} + ... \tag{2.14}$$

By comparing (2.9) and (2.13), we recognise that $z_{m,n} = k_{m,n}a$. This then gives us $k_{m,n}$ in terms of δ,

$$k_{m,n} = \frac{n\pi}{a\delta} + \frac{(4m^2 - 1)\delta}{8\pi an(1+\delta)} + \dots \tag{2.15}$$

Taylor expanding $(1+\delta)^{-1}$ and squaring the resulting expression, we obtain,

$$k_{m,n}^2 = \left(\frac{n\pi}{a\delta}\right)^2 + \left(\frac{m}{a}\right)^2 - \left(\frac{1}{2a}\right)^2 + O(\delta) \tag{2.16}$$

This then gives us our expression for the energy eigenvalues,

$$E_{m,n} = \frac{\hbar^2 k_{m,n}^2}{2m_0} = \frac{\hbar^2}{2m_0}\left[\left(\frac{n\pi}{a\delta}\right)^2 + \left(\frac{m}{a}\right)^2 - \left(\frac{1}{2a}\right)^2\right] + O(\delta) \tag{2.17}$$

3. Spherical Shell

The solution for the eigenfunctions of the spherical shell proceeds in a manner very similar to that for the circular annulus. Therefore, in what follows, we shall mostly highlight the unique aspects of the solution, with the common aspects already outlined in the solution above.

Starting with the TISE expressed in spherical polars,

$$-\frac{\hbar^2}{2m_0}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right] = E\psi \tag{3.1}$$

and substituting the separable solution $\psi(r,\theta,\phi) = R(r)T(\theta)\Phi(\phi)$, we obtain,

$$-\frac{\hbar^2}{2m_0}\left[\frac{T\Phi}{r^2}\frac{d}{dr}\big(r^2R'\big) + \frac{R\Phi}{r^2\sin\theta}\frac{d}{d\theta}\big(\sin\theta T'\big) + \frac{RT}{r^2\sin^2\theta}\Phi''\right] = ERT\Phi \tag{3.2}$$

Expanding the above equation and rearranging like terms,

$$\frac{\sin^2\theta}{R}\frac{d}{dr}\big(r^2R^{'}\big)+r^2k^2\sin^2\theta+\frac{\sin\theta}{T}\frac{d}{d\theta}\big(\sin\theta T^{'}\big)=-\frac{\Phi^{''}}{\Phi}=S \tag{3.3}$$

where $k=\sqrt{2m_0E}/\hbar^2$ as before. Also, since LHS = LHS(r,θ) and RHS = RHS(ϕ), we have that LHS = RHS = S.

Again as before $S=m^2>0$. This again gives us the same final expression for Φ as before,

$$\Phi(\phi)=Ae^{im\phi},\quad \forall m\in\mathbb{Z} \tag{3.4}$$

Working now with the LHS, we collect like terms such that,

$$\frac{1}{R}\frac{d}{dr}\big(r^2R^{'}\big)+k^2r^2=-\frac{1}{T\sin\theta}\frac{d}{d\theta}\big(\sin\theta T^{'}\big)+\frac{m^2}{\sin^2\theta}=S' \tag{3.5}$$

where LHS = LHS(r) = RHS = RHS(θ) = S' which is another separation constant.

Focusing on RHS$(\theta)=S'$, we obtain,

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\big(\sin\theta T^{'}\big)+\left(S'-\frac{m^2}{\sin^2\theta}\right)T=0 \tag{3.6}$$

Changing variables from θ to x where $x=\cos\theta$ and $T(\theta)=X(x)$ by definition, the equation above transforms to the associated Legendre equation,

$$(1-x^2)X^{''}-2xX^{'}+\left(S-\frac{m^2}{1-x^2}\right)X=0 \tag{3.7}$$

Setting $m=0$, the power series solution then is,

$$X=C_0\left[1-\frac{S'x^2}{2!}-\frac{S'(6-S')x^4}{4!}+\dots\right]+C_1\left[x+\frac{(2-S')x^3}{3!}+\frac{(2-S')(12-S')x^5}{5!}+\dots\right] \tag{3.8}$$

where X only converges for $|x|<1$. Since $\theta\in[0,\pi]\Rightarrow x\in[-1,1]$, convergence is only achieved for special values of $S=\ell(\ell+1),\forall\ell\in\mathbb{Z}_0^+$. For $m=0$, the solutions for X are the Legendre polynomials of order ℓ [13],

$$P_\ell(x)=\frac{1}{2^\ell\ell!}d^\ell dx^\ell(x^2-1)^\ell \tag{3.9}$$

and when $m\neq 0$, the solution set for X gives the associated Legendre polynomials[13],

$$P_\ell^m(x)=(1-x^2)^{|m|/2}\frac{d^{|m|}P_\ell}{dx^{|m|}} \tag{3.10}$$

where $x=\cos\theta$ as defined earlier. The fact that the associated Legendre polynomials $P_\ell^m(x)$ vanish for $|m|>\ell$ imposes a restriction on our wavenumbers, such that $|m|\leq\ell$ must be satisfied for any valid wavefunction. We finally arrive at,

$$T(\theta)=P_\ell^m(cos\theta) \tag{3.11}$$

Finally, we move on to the radial part,

$$\frac{1}{R}\frac{d}{dr}\big(r^2R^{'}\big)+k^2r^2-\ell(\ell+1)=0 \tag{3.12}$$

which simplifies to the spherical Bessel equation,

$$r^2\frac{d^2R}{dr^2}+2r\frac{dR}{dr}+k^2r^2-\ell(\ell+1)=0 \tag{3.13}$$

We therefore obtain,

$$R(r)=Cj_\ell(kr)+Dy_\ell(kr) \tag{3.14}$$

where j_ℓ and y_ℓ are spherical Bessel functions of order ℓ of the first and second kind respectively. The spherical Bessel functions are related to the simple Bessel functions as via the following expressions,

$$j_\ell(kr) = \sqrt{\frac{\pi}{2kr}} J_{\ell+\frac{1}{2}}(kr) \qquad y_\ell(kr) = \sqrt{\frac{\pi}{2kr}} Y_{\ell+\frac{1}{2}}(kr) \tag{3.15}$$

Enforcing the radial boundary condition $R(a) = R(b) = 0$, we get a matrix equation similar to the case before,

$$\begin{pmatrix} j_\ell(ka) & y_\ell(ka) \\ j_\ell(kb) & y_\ell(kb) \end{pmatrix} \begin{pmatrix} C \\ D \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{3.16}$$

where for non-trivial C and D, the determinant of the coefficient matrix must vanish,

$$j_\ell(ka) y_\ell(kb) - j_\ell(kb) y_\ell(ka) = 0 \tag{3.17}$$

This then allows us to obtain the following relationship,

$$D = -C \frac{j_\ell(ka)}{y_\ell(ka)} \tag{3.18}$$

Inserting this relation into the general solution for $R(r)$, we obtain,

$$R(r) = \frac{C}{y_\ell(ka)} \Big[y_\ell(ka) j_\ell(kr) - j_\ell(ka) y_\ell(kr) \Big] \tag{3.19}$$

Combining $R(r)$, $T(\theta)$ and $\Phi(\phi)$, the total wavefunction then is,

$$\psi_{n,\ell,m} = C e^{im\phi} P_\ell^m(\cos\theta) \Big[y_\ell(k_{n,\ell} a) j_\ell(k_{n,\ell} r) - j_\ell(k_{n,\ell} a) y_\ell(k_{n,\ell} r) \Big] \tag{3.20}$$

where C is a different normalisation constant.

In the thin shell limit, we obtain,

$$k_{n,\ell}^2 = \left(\frac{n\pi}{a\delta} \right)^2 + \frac{\ell(\ell+1)}{a^2} + O(\delta) \tag{3.21}$$

which leads to energy eigenvalues given by,

$$E_{n,\ell} = \frac{\hbar^2 k_{n,\ell}^2}{2m_0} = \frac{\hbar^2}{2m_0} \left[\left(\frac{n\pi}{a\delta} \right)^2 + \frac{\ell(\ell+1)}{a^2} \right] \tag{3.22}$$

APPENDIX B: CURVATURE

We now compute the expression we obtained in (B.12) of III-B. We begin by noting that,

$$g^{-1}\partial_i g = \partial_i \ln g \tag{0.1}$$

where the determinant of our metric is defined as follows,

$$g = g(\tau = 0)\gamma = g_0\gamma \tag{0.2}$$

This then leads to,

$$g^{-1/2}\partial_\tau g^{1/2}\partial_\tau\psi = \partial_\tau^2\psi + \frac{1}{2}(\partial_\tau \ln\gamma)(\partial_\tau\psi) \tag{0.3}$$

Substituting $\psi = \gamma^{-1/4}\chi$, (0.3) becomes,

$$\partial_\tau^2\psi + \frac{1}{2}(\partial_\tau \ln\gamma)(\partial_\tau\psi) = \partial_\tau^2\gamma^{-1/4}\chi + \frac{1}{2}(\partial_\tau \ln\gamma)(\partial_\tau\gamma^{-1/4}\chi) \tag{0.4}$$

We now expand each term in (0.4) separately. Expanding the first term using the fact that $\gamma^{-5/4}\partial_\tau\gamma = \gamma^{-1/4}\partial_\tau \ln\gamma$, we get,

$$\partial_\tau^2\gamma^{-1/4}\chi = \partial_\tau(\partial_\tau\gamma^{-1/4}\chi) = \partial_\tau\Big(-\frac{\gamma^{-1/4}}{4}(\partial_\tau \ln\gamma)\chi + \gamma^{-1/4}\partial_\tau\chi\Big) \tag{0.5}$$

which after a bit of labour simplifies to,

$$\frac{\gamma^{-1/4}}{16}(\partial_\tau \ln\gamma)^2\chi - \frac{\gamma^{-1/4}}{4}(\partial_\tau^2 \ln\gamma)\chi - \frac{\gamma^{-1/4}}{2}(\partial_\tau \ln\gamma)\partial_\tau\chi + \gamma^{-1/4}\partial_\tau^2\chi \tag{0.6}$$

Doing the same for the second term,

$$\frac{1}{2}(\partial_\tau \ln\gamma)(\partial_\tau\gamma^{-1/4}\chi) = -\frac{\gamma^{-1/4}}{8}(\partial_\tau \ln\gamma)^2\chi + \frac{\gamma^{-1/4}}{2}(\partial_\tau \ln\gamma)\partial_\tau\chi \tag{0.7}$$

Combining (0.6) and (0.7) and collecting like terms we obtain,

$$\partial_\tau^2\psi + \frac{1}{2}(\partial_\tau \ln\gamma)(\partial_\tau\psi) = \gamma^{-1/4}\left[-\frac{1}{16}(\partial_\tau \ln\gamma)^2\chi - \frac{1}{4}(\partial_\tau^2 \ln\gamma)\chi + \partial_\tau^2\chi\right] = \gamma^{-1/4}\Big(\partial_\tau^2\chi + U\chi\Big) \tag{0.8}$$

where U is defined as follows,

$$U = -\frac{1}{16}(\partial_\tau \ln\gamma)^2 - \frac{1}{4}(\partial_\tau^2 \ln\gamma) \tag{0.9}$$

We can simplify U further by recalling that $\gamma = \Big(1 - \frac{\tau}{R_1}\Big)^2\Big(1 - \frac{\tau}{R_2}\Big)^2$. In order to simplify U, we must first differentiate $\ln\gamma$ with respect to τ:

$$\partial_\tau \ln\gamma = -2\Big(\frac{1}{R_1 - \tau} + \frac{1}{R_2 - \tau}\Big) \tag{0.10}$$

which leads us to the second derivative,

$$\partial_\tau^2 \ln\gamma = -2\left[\frac{1}{(R_1 - \tau)^2} + \frac{1}{(R_2 - \tau)^2}\right] \tag{0.11}$$

Using the above and simplifying, we obtain the following for U:

$$U = -\frac{1}{4}\left[\frac{1}{(R_1 - \tau)^2} + \frac{2}{(R_1 - \tau)(R_2 - \tau)} + \frac{1}{(R_2 - \tau)^2}\right] + \frac{1}{2}\left[\frac{1}{(R_1 - \tau)^2} + \frac{1}{(R_2 - \tau)^2}\right] \tag{0.12}$$

from which we obtain the final expression for U,

$$= \frac{1}{4}\Big(\frac{1}{R_1 - \tau} - \frac{1}{R_2 - \tau}\Big)^2 \tag{0.13}$$